

Visual Augmentation of Audio Semantics for Accessibility

Detailed technical report

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Code: <https://github.com/adamgavriely/MscFinalProject>, tag `detector-frozen-2026-10-02`

Abstract

Subtitles carry speech. The other sounds of a video – a siren behind the camera, a dog barking in the next room, glass breaking off screen – reach a deaf or hard-of-hearing viewer only if a captioner chose to write them down, and captioning practice says to leave out any sound whose source can be seen. This project builds and evaluates a system that fills this gap automatically. Given a video, it detects the non-speech, non-music sounds, decides for each one whether its source is already visible, and shows a generated picture of the sound beside the video only while an *off-screen* sound is heard. No model is trained: the system is a chain of open models (two sound detectors, two audio-language models used as listeners, a third audio model used as a witness, a 27-billion-parameter vision-language model for the on-screen decision, and a text-to-image model), joined by rules that were each fixed on a development set before being tested.

The system is scored per sound against human labels on a development set of 71 clips and a test set of 88 clips. A *hit* is a picture of the right kind of sound that starts between 0.5 s before and 1 s after the sound; a *wrong* picture shows a sound whose source is on screen, belongs to another sound, or has no sound under it; the cost per clip is $(4 \cdot \text{missed} + 2 \cdot \text{wrong})/\text{clips}$. The final system finds 29 of 58 needed sounds with 15 wrong pictures on the development set (cost 2.056) and 24 of 65 with 24 wrong on the test set (cost 2.409). The September baseline scored 3.690 and 2.909 on the same sets, and showing nothing at all costs 3.268 and 2.955. The test set was read after every accepted change during development, so the test numbers are a report on a seen set, not a blind confirmation; Section 8 lists this and the other threats to validity.

The largest remaining losses are well located. With every stage made perfect on the same detections, the pipeline would still miss about a fifth of the needed sounds, because no model hears them under speech and music; and the single largest decision loss is the on-screen check, where today’s open vision-language models confuse “the object is in the frame” with “the object is making this sound now”. The development record – about seventy pre-registered rounds of experiments – is given in Appendix E. A study with deaf viewers was not run, so the claim that the pictures help is untested.

Contents

1	Introduction	4
1.1	Motivation	4
1.2	Problem statement	4
1.3	Research questions	4
1.4	Contributions	5
1.5	How to read this report	5
2	Background and related work	6

3	Task definition and metric	7
3.1	What counts as a needed sound	7
3.2	Matching pictures to sounds	7
3.3	Viewer cost	8
3.4	Statistics	9
3.5	Pass rules on the development set	9
4	Benchmark and labels	9
4.1	Data sets	9
4.2	Label quality	10
5	The final system	10
5.1	Overview	11
5.2	Stages at a glance	11
5.3	Stage 4: detection	11
5.3.1	Models	12
5.3.2	Steps in execution order	13
5.3.3	The scene check	14
5.4	What may be drawn	14
5.5	Stage 5: the on-screen gate	15
5.6	Stage 5b and 6: what the picture shows	15
5.7	Stage 6b: display	16
5.8	Scored harness versus live pipeline	16
6	Results	16
6.1	Main result	17
6.2	Sensitivity	17
6.3	Where the misses come from	17
6.4	How far could this pipeline go?	18
6.5	What the on-screen gate is worth	18
6.6	Other measurements	18
7	Error analysis	19
7.1	Failure modes	19
7.2	The central limit: “present” is not “making the sound”	20
7.3	The detector floor	20
7.4	Is the gold too strict?	20
7.5	A worked example	21
8	Disclosures and threats to validity	21
9	Limitations and future work	22
9.1	Limitations	22
9.2	Future work	23
10	Reproduction	23
10.1	Code state	23
10.2	Environment and models	23
10.3	Running one video	24
10.4	Running a folder of clips on the cluster	24
10.5	Reproducing the scores	24
10.6	The Decision Inspector	24
10.7	Where the records are	25

11 Conclusion	25
A Prompts	26
B Configuration of the final system	27
C Sources of the numbers	28
D Internal names used in the development record	29
E Development history	30
E.1 How the data grew	30
E.2 Working rules	31
E.3 Phases	31
E.4 The shipped chain	33
E.5 What was tried and rejected	34
E.6 Pictures	36
E.7 Earlier evaluation by model judge (superseded)	36
E.8 Comparisons between versions	36
E.9 Round-by-round record	37

1 Introduction

1.1 Motivation

Deaf and hard-of-hearing (DHH) viewers receive speech through subtitles and, where available, sign language. Neither channel is built for the rest of the soundtrack. Subtitles for the deaf and hard of hearing (SDH) may name a sound effect in brackets, but the DCMP Captioning Key [12] tells captioners that the source of a sound “may be omitted if it can be clearly seen onscreen”, and that background effects are written only when they matter to the plot. A study of more than 715,000 YouTube videos found hand-written non-speech captions in only 5–7 % of them, and environmental sounds in 2–3 % [31]. Field studies of sound-awareness tools [25, 23, 43] show that DHH users want to know about exactly the sounds that carry information – knocks, alarms, sirens, a baby crying, a dog barking – and interviews on non-speech captions report that viewers want the type and the timing of story-relevant sounds [32].

The information a hearing viewer gets from an ambient sound is therefore lost in proportion to how far its source is from the frame. A siren over a visible ambulance tells the viewer nothing new; a siren over a crowd looking up *is* the scene. This project targets the second case. It differs from audio-to-image generation [44, 6], which turns a whole soundtrack into a picture regardless of the video, and from sound-source localisation [42, 47], which asks where in the frame a sound comes from but not whether the viewer needs to be told about it.

1.2 Problem statement

Given a video, produce the same video with a side panel that shows, while an ambient sound is heard whose source the viewer cannot see, a picture of that sound – and shows nothing otherwise. Speech and music are never drawn (they have their own channels). Human non-word sounds such as laughter, coughing or a scream *are* drawn. No model is trained or fine-tuned: the system is an inference pipeline over open models, and human work is limited to the labels used for evaluation. The system must run on one video at a time (“online”): every rule has to work without seeing other clips, and every expensive step needs a fixed trigger.

Concretely, for every sound in a clip the system must answer four questions:

1. **Is there a sound?** Find each non-speech, non-music sound and when it starts.
2. **What is it?** Name it at a level a picture can show (for example *dog bark*, not *animal*).
3. **Can the viewer already see it?** If the thing making the sound is visibly making it, stay silent.
4. **What should the picture show, and for how long?**

1.3 Research questions

The research questions are those of the project proposal (6 August 2026); Section 11 gives the answer to each.

- RQ1. Which types of audio information contribute most to scene understanding when presented visually?
- RQ2. What level of semantic granularity should be used to represent ambient sounds?
- RQ3. Can automatically generated visual augmentations improve accessibility beyond subtitles?
- RQ4. How should semantic audio augmentation systems be evaluated?
- RQ5. How does the proposed pipeline compare with representative audio-to-visual approaches?

1.4 Contributions

1. **A task definition** with a precise rule for when a picture is due: a sound is *needed* when it is neither visible (“seen making the sound now”) nor obvious (“with the sound off, the viewer would still know it is happening”), and it is an event, not steady background (Section 3).
2. **A per-sound benchmark**: a development set of 71 clips and a test set of 88 clips with timed labels, visibility and importance for every sound (Section 4).
3. **A metric that matches the decision**: hits, wrong pictures split by cause, and a viewer cost that prices a missed sound at twice a wrong picture, with a sensitivity scale for on-screen pictures (Section 3.3).
4. **A working system** of open models with no training, runnable on one video (`main.py`; a ComfyUI wrapper is in `comfyui_nodes/`) (Section 5).
5. **An error analysis** that locates the limit of today’s open models for this task: sounds no model hears, and the confusion between an object being present and the object making the sound (Section 7).
6. **A complete, pre-registered development record**: every idea tried, its pass bar written before the run, its result and the reason it was kept or dropped, including negative results about automatic evaluation with model judges (Appendix E).

1.5 How to read this report

This is a technical report in the sense of the course guidelines: it aims at a depth that lets a reader reproduce the system and its numbers. The main body describes the **final system** only. Section 2 places the work. Sections 3 and 4 define the task, the metric and the data. Section 5 describes the final system stage by stage with every threshold, Section 6 gives the results, Section 7 the error analysis, Section 8 the threats to validity, Section 9 limitations and future work, and Section 10 exact reproduction steps.

The final system was reached through a pre-registered development process of about seventy rounds. That process, the earlier versions of the system and every rejected idea are described in Appendix E; the body refers to it only where a rejected alternative explains a design choice. Appendix C lists the result file behind every number in the body, and Appendix D maps the short internal names used in the development logs to the plain names used here.

Plain-language summary. The system listens to a video and looks at it. When it hears something (for example a dog barking) it asks: can the viewer already see the dog barking? If yes, it does nothing. If no, it draws a picture of a barking dog and shows it next to the video while the bark is heard. On the test videos it finds about 37% of the sounds a deaf viewer would need, and about half of the pictures it shows are right. Most of what it misses, it misses because no current open model hears the sound under speech and music, or because the vision model believes a visible object is the source when it is not.

Names used in this report. *The final system* is the code at git tag `detector-frozen-2026-10-02`, configured by `config.use_shipped()`. *The September baseline* is the system of the September test table, configured by `config.use_scored()`. *Show nothing* is a system that never shows a picture. *The development set* and *the test set* are defined in Section 4; *the screening set* is 415 AudioSet-Strong clips used only to screen audio rules.

Repository map. The code is in `src/` (one package per stage), the entry point is `main.py`, the configuration is `config.py`, the benchmark and scoring harness are in `benchmark/`, cluster job scripts in `slurm/`, this report in `docs/report/`, and the development record in `docs/history/`. The top-level `README.md` gives the full layout.

2 Background and related work

Accessibility of non-speech sound. The DCMF Captioning Key [12] is the practice the visibility rule of this project follows: describe a sound only when the picture does not already tell the viewer. May et al. [31] measured how rarely this happens on YouTube, and a later survey of 168 DHH viewers with interviews [32] found that they want the type, the timing and the story-relevant sounds, and want a choice. An EEG study [39] found that standard sound captions lose emotional information. Beyond Subtitles [1] interviewed 11 DHH viewers about captions and graphics for non-speech sounds and built an authoring tool on sound event detection; it supports showing *important* sounds only. Caption Royale [13] (39 DHH participants) and prosody captions [37] show that visual styling is welcome when it is readable and does not distract. Sound-awareness systems – head-mounted displays [24], home displays [25, 23], SoundWatch [43], ProtoSound [38] – show that DHH users want urgent sounds fast and are worried by false alarms; this is why a wrong picture has a cost in our metric. Earlier work found icons learnable but slower than text at first [49] and peripheral sound visualisations useful [30]. None of this work generates imagery for ambient sound in recorded video, selected automatically and gated by what is already visible.

Sound event detection and tagging. AudioSet [18] (about two million 10-s clips, an ontology of 632 classes of which 527 are released) is the label space of almost every tagger. PANNs [28], AST [20] and BEATs [9] are clip-level taggers; AudioSet-Strong [22] adds labels with 0.1-s timing. Text-queried, open-vocabulary detectors score a free-text name per frame: CLAP [15], FLAM [51], FlexSED [21] and the audio part of PE-AV [48]. Sound event detection is usually scored with event-based precision, recall and F1 with a time collar [33] or with threshold-free PSDS [5, 16]. Our metric borrows the onset-only collar but makes it asymmetric (a picture may start a little early but not late), and reports a cost rather than F1, in the spirit of cost curves [14]. A known weakness of taggers matters here: a quiet sound under speech or music is ranked below *Speech* or *Music*, and most of the sounds in our benchmark are exactly of this kind.

Audio-language models. Qwen-Audio and Qwen2-Audio [10, 11], SALMONN [45] and Qwen3-Omni [53] answer free questions about audio and are benchmarked on MMAU [40]. In this project two of them (Qwen3-Omni and Audio Flamingo Next) are used as *listeners*: they are asked whether a candidate sound is present and to list what they hear, and their answers decide whether a weak detection is kept. Their free text never reaches the picture prompt, because asked “what is making this sound?” the listener named a different maker on 35 of 54 sounds (a hen’s cluck became “goats bleating”; docs/history/preregistrations/freeze_picture_setup_2026-09-25.md).

Vision-language models. Qwen2.5-VL [3], LLaVA [29] and their successors answer open questions about frames and are the only component that can decide whether a sound’s source is visible without a fixed object list. Known failure modes matter for this task: a prior on answer letters in multiple choice [58], weak handling of negation [2], and a tendency to treat a present object as the sounding one, which recent audio-visual benchmarks also report (the project’s VLM panel notes list them, docs/history/review/vlm_panel_2026-10-01/f2_prompts.md). Section 7 shows that this last failure is the main limit of the gate.

Sound-source localisation and off-screen sound. Self-supervised localisation [41, 8, 42] and object-aware localisation [47] point to where in the frame a sound comes from. Juanola et al. [26] show that many of these models barely change their output for silence, noise or off-screen audio, which is the case that matters here; training with such negatives helps [27]. Off-screen separation [34, 46, 56] outputs audio, not a decision about what the viewer needs. These models answer “where?”, not “whether?”, and none was found that decides “visible and making the sound now”; two such signals (an audio-visual localiser and an audio-visual synchrony model) were tested as gate votes and rejected (Appendix E).

Audio-to-image generation. Sound2Scene [44], AudioToken [54], SonicDiffusion [6], Seeing Sound [35] and AudioCanvas [17] draw an image from audio; TempoTokens [55] adapts a text-to-video model to audio. They are the natural comparison family but a poor fit: they condition on the whole clip (and so draw the scene, not the one unseen source), cannot be told to leave an object out, and never look at the video. This project therefore uses text-to-image models (SDXL [36], FLUX.1 [4], finally Qwen-Image [52]) with a short guarded prompt, and isolates the value of the on-screen decision with a *blind* baseline that uses the same detector and generator but draws every sound.

Evaluation with model judges. LLM-as-a-judge has documented position, length and self-preference biases [57]; audio-caption metrics validated against people (FENSE [59], CLAIR-A [50]) exist for text, not for pictures shown beside a video. The proposal planned an automatic protocol (a VLM describes the augmented video, an LLM scores it against a reference). Appendix E.7 reports why it was replaced by per-sound scoring against human labels.

3 Task definition and metric

3.1 What counts as a needed sound

The unit of evaluation is one sound in one clip. For every sound the annotator records a label (mapped to the AudioSet ontology), its onset and offset (about 0.5-s precision), and four judgements (`benchmark/gold/tool_template.html`, `docs/history/plans/metric_per_sound.md`):

visible

the thing making the sound is *seen making it now*. A barking dog on screen is visible; a baby held quietly is not visibly crying; an alarm box on a wall is not visibly ringing. If the source is visible for only part of the sound, tick visible only if it is visible for most of it.

obvious

“with the sound off, would a viewer still know this sound is happening now?” (for example waves when the sea fills the frame).

importance

a property of the sound, rated as if the screen were black: 1 = steady noise of the place, no clear start or end; 2 = an event that can be said in one sentence; 3 = danger or a key story moment (siren, alarm, explosion, gunshot, glass, baby crying, doorbell, phone, horn, scream). When unsure, the lower value.

masked

the sound is covered by speech or music (recorded, not used in scoring).

A sound is **needed** if it is neither visible nor obvious (`benchmark/gold/score_per_sound.py`, `load_gold`), and it is **scored** if its importance is 2 or 3 (`MIN_IMPORTANCE = 2`). A needed sound of importance 1 is “don’t care”: a picture on it is neither a hit nor wrong. Speech and music are never sounds. Every heard sound is labelled, not only the needed ones, because a picture with no labelled sound under it counts as wrong.

Example. In a clip of a street at night, a police siren is heard while the camera shows a crowd looking up. The siren is not visible and not obvious from the picture, and it is importance 3: it is *needed*. If the police car then drives into view with its lights on, a second siren row is labelled visible.

3.2 Matching pictures to sounds

Pictures are scored as they are *displayed*: the scorer replays the compositor’s display rules (repeat merge, minimum time on screen, grouping, at most three slots) before matching (`load_pictures`).

The rules, from `score_clip`:

1. **Onset window.** A picture can match a sound only if the picture *starts* within $[t_{\text{onset}} - 0.5\text{ s}, t_{\text{onset}} + 1.0\text{ s}]$. The end of the picture is not scored. (A scoring panel first proposed +2 s; the author set +1.0 s on 19 Sept, before any per-sound score.)
2. **Same family.** The picture’s label and the sound’s label must be the same, or one must be an ontology ancestor of the other, and both must be below the top level of the ontology (depth ≥ 1): *Vehicle* matches *Car*, but the top-level *Sounds of things* matches nothing.
3. **One to one.** Each sound takes at most one picture (the nearest in time). Extra pictures on a sound already taken are *duplicates*: not credited, not counted as wrong.

Each scored needed sound is a **hit** or a **miss**. Each picture that is not a hit or a duplicate is **wrong**, of one of three kinds:

on screen (“visible”) it matches a sound that is not needed (visible or obvious). The viewer could see it; the picture is redundant.

other sound (“cross”) no same-family sound starts in its window, but some labelled sound is playing or starting there. This includes a second picture of a long sound that started earlier and is still playing – the *strict* rule.

nothing (“phantom”) no labelled sound at all is there.

“Wrong” is therefore strict: it counts unneeded pictures, not only false ones. Section 8 gives a human check of how often a “wrong” picture is in fact a real sound the labels missed.

3.3 Viewer cost

The decision metric is the viewer cost per clip,

$$C = \frac{4 \cdot \# \text{missed} + \beta \cdot \# \text{wrong}}{\# \text{clips}}, \quad \beta = 2, \quad (1)$$

lower is better (`COST_MISS`, `COST_FA` = 4.0, 2.0 in `score_per_sound.py`). A missed needed sound costs twice a wrong picture: one hit pays for at most two wrong pictures, so a picture is worth showing when the chance that it is right exceeds $\beta/(4 + \beta) = 1/3$. The weights come from the rubric of the September judge (a withheld picture scored 0 where one was due, a redundant one about 3 of 4) and were declared for a gate sweep before any per-sound result; they were promoted to a reported outcome only after per-sound F1 came out null (22 Sept), so the choice of $\beta = 2$ is *post hoc* and is reported with sensitivity.

On-screen weight w . A picture of a sound whose source is on screen is less harmful than a picture of nothing. From 1 Oct the cost is also reported as

$$C_w = \frac{4 \cdot \# \text{missed} + w \cdot \# \text{visible} + 2 \cdot \# \text{cross} + 2 \cdot \# \text{phantom}}{\# \text{clips}}, \quad w \in \{0, 1, 2\}, \quad (2)$$

with $w = 2$ (Eq. 1) as the primary. Wins that hold only at $w = 1$ were to go to a separate “more hits” profile, never to the main system.

Why not F1. Per-sound F1 on the same labels was the pre-registered headline from 22 Sept. It weighs a miss and a wrong picture equally, and the gate’s value lies almost entirely in the wrong pictures it removes, so the F1 difference between the gated system and the blind baseline was null in every version (for example +0.059 [−0.030, +0.144], $p = 0.18$, on the September test table) while the number of wrong pictures fell significantly (Section 6.5). F1, precision and recall are still computed by the scorer.

3.4 Statistics

All comparisons are paired by clip. A **paired clip bootstrap** (2000 draws, seed 0) resamples clips and scores every system on the same resampled clips. Two read-outs are used, and the report always says which:

- *Development comparisons*: mean per-clip cost difference d , a 95 % percentile interval, and a **one-sided** p = share of draws with $d \geq 0$. All p -values of the final development phase are of this kind; a two-sided value is about twice as large.
- *September test table*: **two-sided** $p = \min(1, 2 \min(P(d \geq 0), P(d \leq 0)))$ with Holm step-down correction at $\alpha = 0.05$ over two pre-declared families (7 and 3 comparisons).

With 88 test clips the smallest cost difference this design can detect is about 0.19 per clip; a difference of 0.09 would need about 385 clips. Most single changes made during development (Appendix E) are smaller than that, which is why decisions were made on pre-declared rules on the development set and not on test significance.

3.5 Pass rules on the development set

A change to the system was accepted only if it passed a rule written before its run, on the development set:

main rule hits $\geq$ before, wrong $\leq$ before + $2 \times$ (hits gained), cost $<$ before, and no needed hit lost;

fewer-pictures rule cost $<$ before, at least three wrong pictures removed per hit lost, and at most three hits lost;

screening a rule about the audio alone could first be screened on the screening set (Section 4) and stopped there, but never accepted on it.

The test set was read only after a change was accepted, and only reported. Section 8 counts how often that happened.

4 Benchmark and labels

4.1 Data sets

All final numbers are on two sets of short video clips (10–20s) labelled per sound with the fields of Section 3.1:

The development set 71 clips with 58 needed sounds. Every design decision was made on this set.

The test set 88 clips with 65 needed sounds, read after a change was accepted and only reported. Both sets mix three sources: clips of an earlier clip-level benchmark (Wikimedia Commons, the Internet Archive, UnAV-100 [19], VGGSound [7] controls and YouTube, including film scenes); new YouTube clips found by scene type (transport, city walk, market, nature, home, incident, work event, water, film/TV; one 15–20-s cut per upload); and 10-s clips from the AudioSet-Strong evaluation split. The sets grew in two steps, described in Appendix E.1; new clips were assigned to development or test by a hash of the clip name or by balanced pairs, never by looking at results. Most clips are for research use only and are not redistributed.

The last column counts clips by scenario: *unseen* (an ambient sound whose source is off screen), *mixed* (some sources visible, some not), *seen* (every source on screen) and *none* (speech or music only). About half of the clips (33 of 71 and 45 of 88) have no needed sound at all; on them, the best a system can do is show nothing.

Table 1: The two evaluation sets (recomputed from `benchmark/gold/annotations/gold_AG.json` with the scorer’s own loader). “Needed” = needed sounds of importance 2–3, the scored denominator.

set	clips	sounds	needed	imp. 3	clips w. needed	unseen/mixed/seen/none
development set	71	154	58	17	38	24/16/23/8
test set	88	154	65	13	43	21/23/22/22

Three unreachable sounds. The labels keep every non-speech sound, while the final system’s label filter (Section 5.4) never draws three of the development labels (*Clang* in one clip and two *Whack*, *thwack* rows in another). They are misses for every system by construction; without them the development set would have 55 needed sounds.

Other data. Two further sets were used for narrower questions:

The screening set 415 clips from the AudioSet-Strong evaluation split (252 “complex” clips where speech or music covers half the clip and a sound lies under it, 163 random). Its 0.1-s labels need no new annotation, so it was used to screen rules about the audio alone (for example “how often is a detection that no listener names a real sound?”). It never decided a change by itself, because AudioSet labels only a few sounds per clip: a cost computed on it counted most correct detections as false.

DCASE 2025 Task 3 258 events with on/off-screen labels, used once as an outside check of the on-screen decision of an earlier version (66.7 % agreement; the DCASE label is geometric, so it bounds the gate only loosely).

4.2 Label quality

All labels are the author’s. A packet of 30 clips for a second annotator (stratified, half development, half test) was built but no labels came back, so inter-rater agreement is unknown. The only reliability number is intra-rater, on the earlier clip-level labels: 60 clips relabelled blind, 78 % agreement, Cohen’s $\kappa = 0.60$. Three further facts bear on label quality (all listed again in Section 8):

- **Anchoring.** Earlier clips were pre-filled from the system under test, the labelling tool showed detector suggestions on 37 of the 50 newest clips, and the 19 AudioSet-Strong clips were labelled with the AudioSet labels available behind a button (shown on 16 of 19).
- **Visibility re-check.** The author re-judged, blind to the old answer, 14 borderline development sounds where the system and the labels disagreed; 3 changed from seen to needed (a church bell, a fire alarm, a thunder clap). A control round on 20 random development sounds changed none. The test labels were never changed.
- **Missing sounds.** In a blind check of pictures counted as wrong, about 1 in 6 turned out to be a real, wanted, off-screen sound the labels do not list. No label was edited; absolute costs are therefore slightly pessimistic for every system alike.

The labelling guide for the newest clips adds two rules: a sound that stops for more than about 2 s and comes back gets a new row, and a long sound whose visibility changes is split into two rows.

5 The final system

This section describes the final system exactly as the code runs it: the configuration `config.use_shipped()` (`config.py`), frozen on 2 Oct 2026 under git tag `detector-frozen-2026-10-02`. Where the scoring harness and the live pipeline differ, the difference is stated (Section 5.8). Every value below gives its configuration variable; Appendix B lists them all.

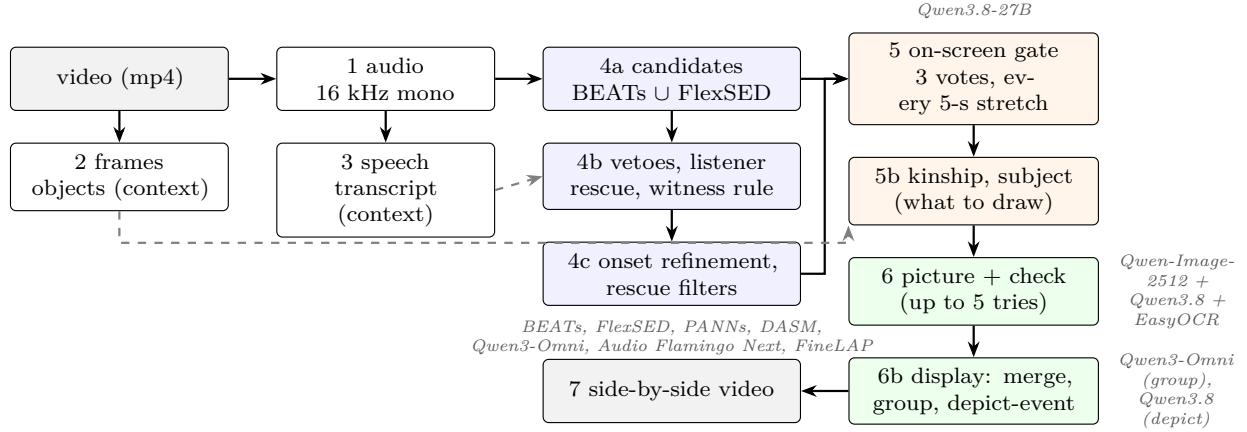


Figure 1: The final pipeline. Blue: detection (stage 4); orange: the on-screen decision (stage 5); green: pictures and display. Dashed: context only (objects and transcript never decide on their own). The scene check of the witness rule in stage 4b also uses Qwen3.8-27B.

5.1 Overview

Figure 1 shows the data flow. In words, for one video:

1. **Hear candidates.** Two sound detectors listen to the whole clip: BEATs (a 527-class tagger) and FlexSED (a detector that finds sounds by name). Anything they hear strongly enough, for at least 0.3 s, becomes a candidate with a start and an end.
2. **Check candidates.** A chain of vetoes removes candidates that only one model claims. Weak sounds that the detectors almost missed get second opinions from two audio-language “listeners” (Qwen3-Omni and Audio Flamingo Next) and a third audio model (DASM).
3. **Witness rule.** A remaining candidate needs a witness: DASM clearly hears it, or both listeners name it, or one listener names it *and* the vision model finds the sound plausible in the scene.
4. **On-screen check (the gate).** A vision-language model (Qwen3.8-27B) looks at the frames around each sound. If the thing making the sound is visibly making it, no picture is shown.
5. **Clean the display.** Repeats close together become one picture; a listener decides whether two same-type pictures up to 8 s apart are one continuing sound; a picture is dropped when its event is visibly happening.
6. **Draw.** A short, guarded text prompt is written for each remaining sound; Qwen-Image draws it; a checker asks whether the picture shows the intended thing and redraws up to five times.
7. **Compose.** The pictures are shown in up to three fixed slots beside the video, each while its sound plays (at least 1.5 s).

5.2 Stages at a glance

5.3 Stage 4: detection

Stage 4 is where most of the development effort went. It turns the audio into a list of timed candidate sounds, each with a family label and a confidence. The entry point is `detect_events` (`src/stage4_audio_event_detection/__init__.py`); unions and vetoes are in `fuse_flexsed`, the filters on rescued spans in `filter_rescued`.

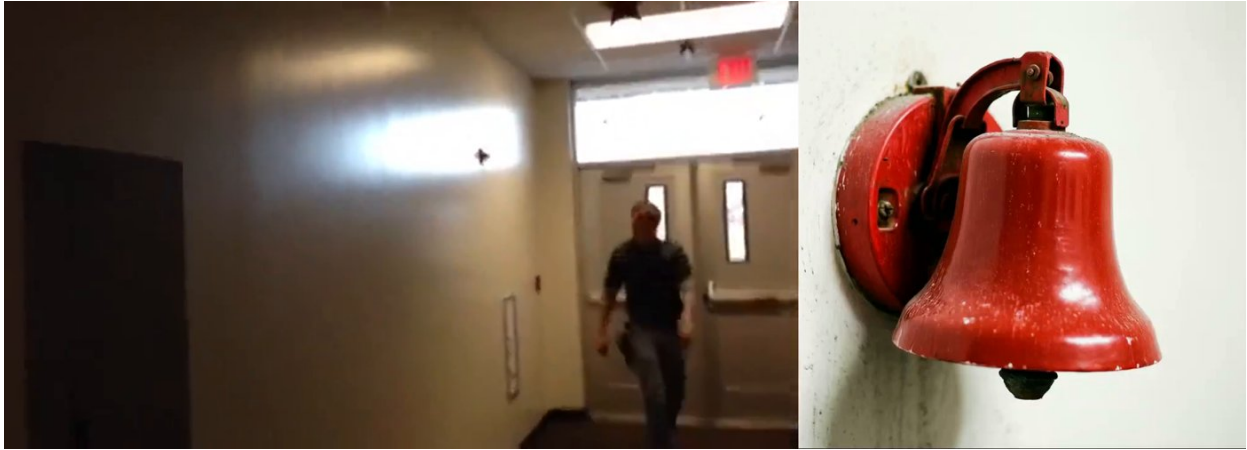


Figure 2: Output of the final system on a development clip (AudioSet-Strong, a corridor). A fire alarm rings from 9.0s; the alarm bell itself is not in the frame. The system detects *Alarm*, the on-screen check finds no visible source, and the panel on the right shows the generated picture from 8.9s while the alarm sounds. In the scoring this is a hit.

5.3.1 Models

BEATs

iter3+ fine-tuned on AudioSet-2M (checkpoint `BEATs_iter3_plus_AS2M_finetuned_on_AS2M_cpt2`). It scores a whole window, so it is run on 2-s windows every 0.25 s; the first two seconds are mirror-padded so a sound at the very start can be heard; each window’s score is stamped 0.5 s before the window ends, so a detected start is never earlier than the true one.

FlexSED

an open-vocabulary, frame-level detector [21]: a self-supervised audio encoder plus a CLAP text encoder (`laion/clap-htsat-unfused`), trained on AudioSet-Strong. It is asked one plain query per family over the 215 *depictable* families (`benchmark/gold/depictable_vocab.json`), 25 frames per second. It hears quiet sounds under speech and music far better than BEATs.

PANNs CNN14

(sound-event version, 32 kHz) used only as a clip-level veto.

DASM

Detect Any Sound Model (ACM MM 2025; weights `CPF2/detect_any_sound`), queried with “sound of <family>” over the same 215 families, 50 frames per second. Used as a witness and a vote.

Listeners

`Qwen/Qwen3-Omni-30B-A3B-Instruct` and `nvidia/audio-flamingo-next-hf`. Each is asked a yes/no question (“Is the sound of <family> present in this recording?”, scored as the logit of *yes* minus the logit of *no*) and an open inventory (“List every distinct non-speech sound you hear in this recording, one per line, most prominent first.”). An inventory line names a family if it matches its name, an AudioSet synonym, a parent or a child, or else has sentence-embedding cosine > 0.6 (`all-mpnet-base-v2`). The audio cut is the candidate span ± 1 s, at least 1 s long.

FineLAP

(`AndreasXi/FineLAP`, MIT) a frame-level audio–text model used as a last veto on rescued sounds.

Table 2: Stages, models and key parameters of the final system (in the order of `src/pipeline.py`).

#	stage	model / tool	key parameters
1	audio	ffmpeg	mono WAV, 16 kHz (<code>SAMPLE_RATE</code>)
2	frames, objects	OWLv2 (<code>owlv2-base-patch16-ensemble</code>)	6 frames, threshold 0.20; context only, never silences a sound
3	speech	faster-whisper base , int8	transcript used only to ask whether people react to a sound; never drawn
4	sound events	BEATs iter3+ AS2M; FlexSED; PANNs CNN14; DASM; Qwen3-Omni-30B-A3B; Audio Flamingo Next; FineLAP	16 ordered steps, Table 3
5	on-screen gate	Qwen/Qwen3.8-27B, greedy, thinking off	6 frames per ≤ 5 -s stretch, 3 votes, majority; silenced only if every stretch is visible
5b	subject	Qwen3.8-27B + word lists	2–6 word phrase; speech text never used
6	picture	Qwen/Qwen-Image-2512	1024^2 , 50 steps, true CFG 4.0; check-and-redraw ≤ 5 tries
6b	display	Qwen3-Omni (group), Qwen3.8 (depict)	merge gap 2.5 s, min. 1.5 s on screen, ≤ 1 s past the end, group ≤ 8 s, 3 slots
7	composite	ffmpeg libx264	720-px panel, 25 fps

5.3.2 Steps in execution order

Table 3 gives the 16 steps. Two words are used throughout: a *rescued* span is one added by a listener after the detectors missed it; a *twinned* span is one both detectors raised.

Table 3: Stage 4 of the final system, step by step.

#	step	rule	variable(s)
1	BEATs spans	a span of any class where the window score ≥ 0.175 ; kept if ≥ 0.3 s long; confidence = peak	AED_THRESHOLD 0.175, AED_MIN_DUR 0.3
2	FlexSED spans	same extractor on FlexSED frames at bar 0.8, ≥ 0.3 s	FLEXSED_BAR 0.8
3	twin union	a FlexSED span with a same-family BEATs span within 1 s is merged into it (earlier start kept; confidence = the stronger side, rescaled); otherwise kept as FlexSED-only	TWIN_MAX, UNION_START min
4	mirror veto + listener keep	a BEATs-only span is dropped if FlexSED’s top query in its window is another family ≥ 0.7 while its own family is < 0.4 ; put back if the Qwen listener accepts it and an inventory names it	MIRROR_VETO 0.7, KEEP_NEEDS_V4
5	masked-weak veto	a BEATs-only span with confidence < 0.5 under speech or music (≥ 0.3) is dropped unless a listener accepts it	MASKED_WEAK_VETO
6	DASM clip veto	a span whose family DASM never reaches 0.0839 anywhere in the clip is dropped unless a listener keeps it	DASM_CLIP_VETO
7	witness rule	pass if DASM ≥ 0.35 for the family within the span ± 0.5 s; else pass if <i>both</i> listener inventories name the family; if exactly one does, ask the scene check (Section 5.3.3); otherwise drop	DASM_LOCAL_VETO 0.35, DASM_LOCAL_KEEP both, SCENE_FIT_LOGIT

#	step	rule	variable(s)
8	inventory keep	a span with confidence ≥ 0.35 for which both inventories exist and neither names its family (or a descendant) is dropped, unless DASM ≥ 0.575 within ± 0.5 s	KEEP_NEEDS_V4_ALL onto, ..._DASM_KEEP
9	band-twin pull	the start moves back to an earlier same-family FlexSED run (≥ 0.5) that ends ≤ 1 s before and starts ≤ 1.5 s before it	BAND_TWIN_PULL 0.5
10	continuation veto	a span is dropped when a same-family FlexSED run (≥ 0.5) began ≥ 1.5 s earlier and is still on at its start (it is a later piece of a sound already shown)	CONTINUATION_VETO 0.5
11	FlexSED veto	a span whose family FlexSED never reaches 0.3 in the clip is dropped	FLEXSED_VETO 0.3
12	PANNs veto	a FlexSED-only span whose family PANNs never reaches 0.05 is dropped unless a listener accepts it	PANNS_VETO 0.05
13	listener band rescue	a FlexSED run with peak in $[0.5, 0.8)$ and no same-family span is added if accepted: peak ≥ 0.6 needs the Qwen inventory; < 0.6 needs both inventories	LISTENER_RESCUE, LISTENER_RULE TIER
14	DASM rescue	a DASM run (≥ 0.575) with no detector span nearby is added if both inventories name the family and the family is new in the clip	DASM_RESCUE, ..._NEW_ONLY
15	onset refinement	for BEATs spans: silence the first t seconds of the first firing window for 26 values of t and take the first t at which 10 % of the class evidence is gone; never earlier than the detected start	ONSET_CAM, ONSET_MONOTONE
16	rescued-span filters	on rescued spans only: FineLAP ≥ 0.329 ; DASM ≥ 0.575 within ± 0.5 s; at most one rescued span per family per clip (the earliest)	FINELAP_VETO 0.329, LISTENER_DASM_VOTE, LISTENER_ONCE

The DASM thresholds 0.0839 and 0.575 and the FineLAP bar 0.329 were calibrated during development (Appendix E); every other threshold was set once and not tuned on the test set. Steps 4–8 and 12–16 depend on listener answers that are computed from a first pass with the September baseline (the candidate “pools”), so the final detector cannot run without that pass (Section 10).

5.3.3 The scene check

When exactly one listener names a weak sound, the witness rule asks the vision model whether the sound is plausible in the scene. The span is cut into stretches of about 5 s; for each, six frames from 1 s before to 1 s after are shown with the question “Could the sound of $\langle \text{label} \rangle$ plausibly be heard in this scene? Answer yes or no.” and its negated twin. The answer is read from the first output token as $d = m(Q) - m(\bar{Q})$, where $m = \max$ logit over {yes, Yes} minus $\max$ logit over {no, No}; $d > 0$ counts as yes, and the sound passes if most stretches say yes. Reading the logits replaced an earlier text answer that was cut off after four tokens on 10–15 % of calls (Appendix E).

5.4 What may be drawn

The label filter (LABEL_FILTER = “depictable”, src/labels.py, is_salient_nonspeech) maps every label to its family and removes: speech with words and its descendants (Speech, Conversation, Narration, Hubbub, Whispering, ...), music and singing and their descendants, the “channel, environment and background” branch, silence and generic labels, textures (Breathing, Rumble, Hum, Whir, Rustle, Wind and its descendants), and labels with no clear source (Sound effect, Onomatopoeia, generic impacts). Everything else is drawable, including non-word vocal events

(shout, scream, laughter, cough, snoring). FlexSED and DASM are asked only about the 215 drawable families.

5.5 Stage 5: the on-screen gate

Stage 5 first groups the detections by family at the display bar 0.35 (`DISPLAY_THRESHOLD`); firings of one family less than 1 s apart are one burst. It then asks Qwen3.8-27B, with greedy decoding and thinking off, for every burst:

1. The burst is cut into $k = \max(1, \text{round}(\text{length}/5))$ stretches; each stretch gets six frames, evenly from 1 s before to 1 s after it.
2. Three votes per stretch (prompts in Appendix A):
 - name** “Name the thing in these frames that is making that sound ... or answer exactly: nothing.” A name that shares a word with the label counts yes; otherwise a text-only question asks whether that thing could make the sound.
 - a/b** “(a) you can SEE $\langle \text{label} \rangle$ happening on screen – the source is in the frames and visibly making that sound; (b) $\langle \text{label} \rangle$ is not visibly happening.” Asked in both option orders; if the two answers disagree the vote abstains.
 - describe** a 25-word description of the frames; yes if it names the sound or the named thing, else the same text-only question.
3. A stretch is visible if yes-votes outnumber no-votes. The sound is silenced only if *every* stretch is visible; otherwise its picture is kept for the whole burst.
4. **Kinship.** A sound is also silenced if a silenced sound of the same source overlaps it in time (live: only when the visible label is the same or more specific; Section 5.8).
5. **Speech context.** If at least four words of speech fall within 3 s of a sound, a text-only question asks whether the people react to it; a yes gives the sound priority for a slot and can rescue a sound just below the display bar. Speech never enters a picture.
6. **Disambiguation.** Two different labels with starts and ends within 0.6 s are one sound with two names; the frames are shown with “which is it, (a) A or (b) B or (c) cannot tell?” in both orders.

The rule “silence only if every stretch is visible” leans toward showing: a sound whose source leaves the frame for one stretch keeps its picture.

5.6 Stage 5b and 6: what the picture shows

Subject. The picture prompt is built from the detector’s label, not from free text. A resolve step asks the VLM for at most two qualifier words plus the head word of the source (accepted only if it ends with the head word, names no person and no forbidden kind), then a 3–6-word event phrase. Word-list guards remove persons, places and makers from other ontology branches. Action labels are drawn as their maker (*bark* $\rightarrow$ a barking dog) from a hand table, ontology senses (*Honk* = goose), or a closed VLM choice on the frames. Sounds that are best shown as a word (*Whoosh*, *Thud*, *Bang*, *Smack*, *Whack*) get a fixed comic burst card, and a few sounds a fixed template (thunder = “one large lightning bolt striking down from a dark storm cloud”).

Generation. The prompt is the subject plus a fixed tail (“the whole thing fully in frame, caught at the moment it makes the sound with its visible effect, one large subject filling the picture, plain white background”) and a negative prompt (scenery, landscape, room interior, street, buildings, text, letters, watermark). Qwen-Image-2512 draws it at 1024×1024 with 50 steps and true CFG 4.0; the seed is a hash of clip, label and start time, so a run is repeatable.

Check and redraw. The same Qwen3.8-27B is shown the picture with a shuffled multiple-choice question (the intended thing, known look-alikes, generic confusions such as “a person” or “a building”, “something else”); EasyOCR rejects a picture that contains a readable word. A failed picture is redrawn with a new seed and the confusing thing added to the negative prompt, up to five tries; if all fail, a word card with the sound’s name is shown. The check was validated on known cases (7 of 7 known-bad pictures caught, 33 of 33 deliberately wrong ones refused, 3 of 68 good ones refused, 113 of 115 answers unchanged on reshuffling) but never calibrated against a human rater (Section 8).

5.7 Stage 6b: display

1. **Depict-event:** a picture is dropped if, on six frames around its start, the VLM says its event is visibly happening (and its negated twin agrees) *and* a maker the gate named could be mistaken for the sound.
2. **Repeat merge:** bursts of one label join into one appearance if the gap from the previous real end is ≤ 2.5 s (MERGE_GAP).
3. **Dwell:** each appearance stays at least 1.5 s (MIN_DWELL) and at most 1 s past the real end of the sound (MAX_AFTER_END).
4. **Grouping:** for two same-family pictures whose gap is at most 8 s, Qwen3-Omni hears the audio from just before the first ends to just after the second starts and is asked whether the second is “the same continuing sound ... (one alarm or engine with a short pause)” or “a new, separate event (... a second bark, knock or shot)”; both option orders must say “same” to merge.
5. **Slots:** at most three pictures at once, each in a fixed slot; if more overlap, sounds people react to come first, then confidence. A slot is empty when its sound is silent.

5.8 Scored harness versus live pipeline

The numbers in this report come from the scoring harness, which runs the pipeline’s own stage-4 functions on cached model scores and replays the gate answers of a full run. A parity check (benchmark/gold/parity_check.py) and an independent re-export of every decision (the Decision Inspector, Section 10) reproduce the published counts exactly (0 of 159 clips differ). The live configuration use_shipped() differs from the scored one in four known ways:

- **Kinship direction.** Live: a visible label silences only the same or a more specific label. Scored: either direction. This was a deliberate logic fix made after the scored tables; its effect on the final numbers was not re-measured.
- **Picture end.** Live pictures end at most 1 s after their sound; scored pictures had no cap. Starts – the scored quantity – are identical; on 3 TEST clips a picture ends 0.04–0.38 s earlier live.
- **Subject path.** The scored runs used the older depiction path, which can drop a sound whose depiction fails twice and feeds a similarity-based merge; the live path never drops a sound at this step. No end-to-end equality test of main.py against the scored numbers exists.
- **Pictures.** Scored runs used placeholder or FLUX.1-schnell pictures; the live system uses Qwen-Image with the check. Pictures are not part of the per-sound score.

6 Results

All numbers are raw counts from the scorer or values derived from those counts with Eq. 1 and 2; derived values are marked. The counts of the final system are reproduced exactly by an independent re-export of every decision (Section 10). Appendix C gives the result file behind each number.

6.1 Main result

Table 4: Main result. Recall = hits / needed; precision = hits / (hits + wrong); “show nothing” costs $4 \cdot \text{needed}/\text{clips}$ (all three derived).

system	development set (71 clips, 58 needed)				test set (88 clips, 65 needed)			
	hits	wrong (v/c/p)	cost	recall / prec.	hits	wrong (v/c/p)	cost	recall / prec.
show nothing	0	0	3.268	–	0	0	2.955	–
September baseline	18	51 (5/37/9)	3.690	.31 / .26	21	40 (2/31/7)	2.909	.32 / .34
final system	29	15 (6/7/2)	2.056	.50 / .66	24	24 (4/15/5)	2.409	.37 / .50

v / c / p = wrong pictures of an on-screen sound / of another sound / with nothing there.

- On the development set the final system finds half of the needed sounds, and two of three of its pictures are right. On the test set it finds 37% and half of its pictures are right.
- The September baseline was *worse than showing nothing* on the development set (3.690 against 3.268, and still 3.521 with a longer repeat merge) and only slightly better on the test set (2.909 against 2.955). The final system is clearly cheaper than showing nothing on both (2.056 and 2.409).
- Against the baseline, the final system has 11 more hits and 36 fewer wrong pictures on the development set, and 3 more hits and 16 fewer wrong pictures on the test set (cost -1.634 and -0.500 per clip). About a third of the development gain carries over to the test set; since many variants were compared on the development set, part of its gain is expected to be selection luck, and the test numbers are the better estimate for new videos.
- On the test set most wrong pictures belong to another sound (15 of 24); pictures of on-screen sounds are few (4).

Significance. The paired bootstrap of the final system against the September baseline on the test set has not been run: it needs per-clip rows that are stored on the cluster, and the baseline’s rows were scored with a shorter repeat merge (2.0s instead of 2.5s), which raises its wrong-picture count. An intermediate version of the final system, without its last display, grouping and witness rules, was cheaper than the baseline on the test set by -0.341 [$-0.636, -0.045$] per clip (one-sided $p = 0.017$, about 0.03 two-sided); the final system has a lower cost than that version on both sets. Appendix E gives every intermediate comparison.

6.2 Sensitivity

On-screen weight. With Eq. 2 the final system costs 1.887 / 1.972 / 2.056 on the development set and 2.318 / 2.364 / 2.409 on the test set at $w = 0/1/2$ (derived). For every accepted change from the baseline to the final system the cost falls at every w , so no decision depends on how on-screen pictures are weighted.

Cost weights. The weights 4 and 2 were fixed before the final development phase and not varied for the final system. On the September test table (Section 6.5) the gated system was the cheapest of all systems for any wrong-picture weight β between 0.39 and 2.56, which contains $\beta = 2$.

6.3 Where the misses come from

For every missed needed sound, the logged decision trail records the step at which it was lost:

Most misses are not “never heard”: the sound was a candidate and a later rule removed it, most often because the listeners did not confirm a weak detection. The on-screen gate silences few needed sounds (5 and 3), but it is also why the on-screen wrong pictures remain (6 and 4).

Table 5: Fate of the missed needed sounds of the final system.

	dev.	test	where
misses	29	41	
lost at a named step	20	32	development: listener band rescue 6, on-screen gate 5, DASM vote 2, family merge 2, mirror veto 2, scene check 1, masked-weak veto 1, one rescue per family 1. Test: band rescue 6, continuation veto 5, family merge 4, gate 3, mirror veto 3, DASM vote 2, BEATs extraction 2, seven other steps 1 each
never heard by any detector	6	6	no span of the family near the onset
timing / matching	3	3	a same-family span exists but starts outside the window

6.4 How far could this pipeline go?

A ceiling analysis replaces one stage at a time by the gold answer on the *same* detections. It was run on the saved pictures of an earlier version (25 of 55 needed sounds, 27 wrong, cost 2.451 on the development set; the final system adds display and witness rules on top), so its absolute numbers belong to that version; its proportions are the best available map.

Table 6: Single-stage oracles (development set, earlier version, 55 needed).

oracle (one stage made perfect)	hits	wrong (v/c/p)	cost	Δ
none (version as run)	25	27 (9/16/2)	2.451	–
perfect on-screen gate	30	18 (0/16/2)	1.915	–0.536
perfect listener, filters lifted	34	24 (7/15/2)	1.859	–0.592
perfect vetoes / filters	30	27 (9/16/2)	2.169	–0.282
perfect timing	26	20 (11/7/2)	2.197	–0.254
perfect family	25	26 (14/10/2)	2.423	–0.028
all stages perfect (ceiling)	43	3 (0/1/2)	0.761	–1.690

Applied in sequence, the gate gives the largest single step (–0.676). Even with every stage perfect, 12 of 55 needed sounds stay missed: 5 are heard by no model at all (a hammer, an explosion, a clang, two golf whacks) and 7 are too faint or too badly timed. This is the *detector floor*: with today’s detectors, no decision layer can find more than about four in five needed sounds.

6.5 What the on-screen gate is worth

The value of the gate itself was measured against a *blind* system (same detector and picture model, no gate) on the September test table (60 test clips, the baseline’s detector, the same gate model and rule):

The gate removes about half a wrong picture per clip at the price of a small loss in recall. Its value lies almost entirely in what it does *not* show, which is why F1 – which weighs a miss and a wrong picture equally – does not move. This comparison was not repeated with the final detector.

6.6 Other measurements

Gate accuracy. On the labelled development sounds the gate has balanced accuracy 0.617: it silences 16 of 43 on-screen sounds and keeps 31 of 36 needed ones. A smaller model of the same family (Qwen2.5-VL-7B) reached 0.61. Shifting the sampled frames by 0.5 s changes 6 of 79 verdicts (7.6 %).

Table 7: Gated minus blind, September test table; two-sided p with Holm correction.

measure	gated – blind	95 % interval	Holm p
F1 (pre-registered primary)	+0.059	[−0.030, +0.144]	0.18 (unadjusted, null)
precision	+0.137		0.010
wrong pictures per clip	−0.517	[−0.800, −0.283]	< 0.001
viewer cost per clip	−0.833		0.012
clean-clip accuracy	+0.219		0.006
unseen clips: gated vs show nothing	−3.077		< 0.001
all clips: gated vs show nothing	−0.233		0.585

Listener agreement as evidence. On the screening set, a detection that neither listener names is a real sound 22 % of the time, one named by one listener 37 %, and one named by both 62 %. This is the basis of the witness rule. A listener asked about the whole clip proposes extra sounds that are real 80 % of the time (74 of 92), against 38–39 % for the raw detectors; they still become wrong pictures in the full pipeline, because the gate passes off-screen ones whose name is wrong, or because the labels list only the salient sounds.

Onset timing. On 255 DCASE 2025 Task 3 events, the onset method places 80 % of the detected events within 0.5 s (mean absolute error 0.35 s). Only 26 % of those short-clip events reach the display bar at all, which again points to detection, not timing, as the limit.

Pictures. In a blind glance test by the author (54 sounds, 1.5 s per picture, no name shown), Qwen-Image pictures were recognised 26 times against 14 for the earlier FLUX.1 pictures with the same text (+0.222 [+0.074, +0.370]). One of the 26 sent a false message (a telephone bell drawn as a desk bell).

7 Error analysis

This section explains *why* the remaining errors happen, using the per-item error lists, an audit of the on-screen gate, two expert review panels and the logged decision trail of every clip (sources in Appendix C).

7.1 Failure modes

1. **Presence taken for production.** The gate’s commonest error: an object being in the frame is taken as that object making the sound now. Examples from the development set: a church tower is visible, so the ringing bell is silenced although nothing in the frames shows it ringing; a golfer stands over the ball, so the whack that happens later is called visible; perched macaws are taken as the source of a bird call made off screen. In the audit this class explains 23 of 36 gate errors across development and test. Two of the three votes effectively measure presence (“name the thing” echoes the label on 30 of 54 development “seen” stretches, and the follow-up “could that thing make it?” is answered from world knowledge), so the majority vote measures presence too.
2. **The “nothing” escape.** The naming question allows “nothing”, and the VLM answers it on 42 % of the stretches the gold calls seen (32 of 76): the maker is small, hidden, or its action is fine (a hand, a coin), the camera is the walker (footsteps), or only the rain is visible during thunder. These sounds keep their picture; this explains 32 of the 38 development gate misses (wrongly kept on-screen sounds).

3. **Masking.** Needed sounds under speech or music are scored low by BEATs, whose top labels there are *Speech* or *Music*. FlexSED hears more of them but mostly below its bar; the listener rescue recovers some, at a price in wrong pictures (Appendix E).
4. **Cross pictures.** On the test set 15 of 24 wrong pictures belong to another sound. Examples from the development ledger: a gunshot picture over footsteps, a crowd picture over laughter, a scream picture over a bird call. These are mostly confident detector errors with a plausible scene, which neither the witness rule nor the gate can refuse.
5. **Timing and repeats.** A sound that pauses for more than 2s and returns is a new labelled event; the system either merges it (a miss) or draws a late second picture of a long sound (a cross). Neither the listener nor the detectors can tell a 2-s pause from a dropout.
6. **Read-out artefacts.** Asked to answer “(a)” or “(b)”, the 27B model favours a position (“(a)” on 88% of answers in one test); asked yes/no, it has a “no” habit; asked for free text, its answer can be cut off. The fix used in the final system is to read the yes/no logits of the first token, minus those of a negated twin question. This removes the artefacts but not the underlying weakness: as a gate signal the logit margin reaches AUROC 0.647, under the pre-set 0.65 bar.

7.2 The central limit: “present” is not “making the sound”

Every attempt to improve the gate in the project landed on the same trade-off line. In September an OWLv2 vote, a SAM 3 vote and a looser silence rule each removed exactly two wrong pictures for every real needed sound they lost, which is break-even at $\beta = 2$. In the October rounds a box-and-crop check, an audio-visual synchrony model, an audio-visual localiser, a human-style “name, find, zoom, ask” chain and a larger or different VLM each silenced more on-screen sounds only by silencing about as many needed ones. A model swap from Qwen2.5-VL-7B to Qwen3.8-27B moved balanced accuracy from 0.61 to 0.62, and Gemma-4-31B moved along the same line. The VLM audit concluded “not the model”: what is asked of the frames – *is this object making this sound right now?* – is a question about an action at an instant, and still frames from open VLMs verify presence well and action poorly.

Two observations make the limit concrete. First, the gold’s own notion of *obvious* encodes inference the gate does not do: a bell in a tower is *not* obvious (one cannot see it ring), thunder under visible rain *is*. A rule that treats “concealed actions” as not visible passed the development test (30 / 14 / 1.972) but in practice fired only on bells, and was not accepted as a general rule. Second, a larger vision model (Qwen3-VL-235B) was the only untested lever named by every panel; it does not fit the available disk and is left as future work. The audit’s own reading is that a larger model should be tested on the presence-versus-production question, not assumed to fix it.

7.3 The detector floor

The ceiling analysis (Section 6.4) puts a hard limit on any decision layer: about one needed sound in eleven is heard by no model at all, and about one in eight more is too faint or too mistimed to recover. Every attempt to hear more of them (Appendix E) found real sounds – 80% of what the best listener chain proposed on the screening set was real – but brought two to four wrong pictures per new hit on the video benchmark. The reason is the combination of the two limits: a newly heard sound still has to pass the gate, and the gate cannot tell a right off-screen sound from a wrong one.

7.4 Is the gold too strict?

Two human checks bound the effect of the strict “wrong” rule. In one check the author rated, blind, samples of “wrong” pictures: about one in six (2/11 and 3/20) is a real, wanted, off-screen sound the gold does not list. In a second check, 12 current wrong pictures were rated: 4 not heard,

5 real but with the source on screen, 2 real and off-screen but timed away from the gold row, 1 unsure; no clean case for a gold edit was found. The ceiling analysis adds that the 14 borderline gate-side items sit one judgement away from the other class: flipping all of them would bring the earlier version’s cost from 2.451 to about 1.9 with no change to the system. The absolute costs are therefore somewhat pessimistic for every system; the comparisons between systems are not affected, because the same gold scores all of them.

7.5 A worked example

Clip `bell_miami` (development) has one needed sound: a *Bell* from 0.2 to 14.5s, importance 3, labelled not visible. Its logged decision trail reads as follows. Stage 4 keeps several *Bell* and *Church bell* candidates. The gate cuts the 14-s burst into three stretches of about 4.8s. In each, the name vote answers “church bell” (it shares the word *bell* with the label: yes), the a/b vote says not visibly happening (no), and the description vote says yes. Two of three in every stretch: the sound is silenced, and the needed bell is a miss lost at the gate. The frames show the church; they do not show the bell swinging. This single sound is the whole development gain of the rejected bell rule (“a concealed action is not visible”), and a clean example of presence taken for production: the one vote that asks about the action got it right, and was outvoted by two votes that measure presence.

8 Disclosures and threats to validity

This section collects in one place everything a reader needs to weigh the numbers. Each item says what happened and what it means for the claims.

- D1. The test set is a seen set.** The test set (88 clips) was read after every accepted change – by count about 19 times between 30 Sept and 2 Oct (one pre-registered read, one read after each of 13 accepted changes, and reads of five candidates that passed the development rule but were not accepted), plus re-reads of saved test pictures for sensitivity tables. The 60 older clips of the test set had already been read 11 times before the merge. No choice was made to improve a test number, with two exceptions that were undone (next item); still, choosing among many development variants and reading the test set after each makes the test numbers optimistic. They are a report on a seen set, not a blind confirmation.
- D2. Two test-informed choices, both undone.** A 4-s limit for the grouping rule was set after reading the test result of the unlimited version and was reverted to the pre-registered 8 s; a picture-confidence floor suggested by a test read was not adopted. Rejections of three development passes (two “expected sound” rules and a second-listener rule) cited their test reads alongside screening-set and development reasons.
- D3. One-sided p -values.** All p -values of the final development phase are one-sided bootstrap shares (for example 0.034 one-sided is about 0.07 two-sided). The direct test of the final system against the September baseline has not been run (Section 6.1).
- D4. Split overlap in September.** A second split file used by the stage-level detector bench shared 35 of its development clips with the 60 older test clips, and the FlexSED bar 0.8 was chosen on that split. On the clean development clips the 0.8 bar passes two of three adoption rules and misses the third by 0.011.
- D5. One annotator, two visibility re-checks.** Every label is the author’s; no inter-rater agreement exists. The author re-checked visibility twice on the development set (3 of 14 disputed sounds changed, 0 of 20 random controls changed after a correction); the re-check targeted sounds where the pipeline and the gold disagreed, so it can favour the pipeline. Test labels were never changed.
- D6. AudioSet clips labelled with AudioSet labels visible.** The 19 AudioSet-Strong clips (across both sets) were labelled with the AudioSet labels available behind a button (shown on

16 of 19), and DASM’s checkpoint was selected on that AudioSet split. The labelling tool showed detector suggestions on 37 of the 50 newest clips, and older clips were pre-filled from the system under test.

- D7. “Wrong” is strict.** A wrong picture includes redundant pictures of on-screen sounds and late second pictures of a long sound, not only false ones. A blind check found that about 1 in 6 “wrong” pictures is a real, wanted, off-screen sound missing from the gold. Absolute costs are pessimistic for every system.
- D8. The cost weights are post hoc.** $\beta = 2$ was declared for a gate sweep in September but became the reported metric only after per-sound F1 came out null; it was assumed, not measured with viewers.
- D9. Scored and live configurations differ in four known ways** (kinship direction, picture end cap, subject path, picture model; Section 5.8). The kinship fix in particular was not re-scored.
- D10. Three development sounds are unreachable** by construction (labels the filter never draws), and the ceiling analysis was run on an earlier version of the final system.
- D11. The picture checker is not calibrated against a person.** Earlier panel rulings said no automatic picture checker should reject pictures until it matched the author’s ratings; the final check-and-redraw loop was validated on known cases only and uses the same VLM that writes the subject. The planned sealed human confirmation of the pictures was cancelled. The picture claim rests on one blind glance test by the author.
- D12. The author is also the rater** of the glance test and the missing-sound checks. A second annotator packet exists but was not returned.
- D13. No study with deaf or hard-of-hearing viewers was run.** Every claim about benefit to viewers is modelled through the cost weights.
- D14. Larger vision model not tested.** Qwen3-VL-235B, the one lever every panel named for the gate, did not fit the available disk and is future work only.
- D15. Inconsistent records found while writing this report.** The pre-registration log is long (546 KB) and was written under time pressure; the fact-check for this report found small inconsistencies, none of which changes an accepted change: two records of an intermediate version’s test row (23/28 and 22/26, both cost 2.545, from before and after the grouping limit was reverted), two records of its development wrong split (6/9/3 and 6/10/2, same total), a stale configuration comment (“ ≤ 4 s” for an 8-s value), and the thesis chapter drafts of 27 Sept, which describe an earlier system and are superseded by this report.

9 Limitations and future work

9.1 Limitations

- **Small benchmark.** 71 development and 88 test clips; one or two pictures decide a development comparison, and the test set can detect only cost differences of about 0.19 per clip.
- **One annotator; no viewer study** (Section 8, D5, D13).
- **Detector floor.** About one needed sound in eleven is heard by no model, mostly under speech or music.
- **Gate limit.** Balanced accuracy about 0.62; presence is taken for production; “nothing” is used as an escape.

- **Runtime.** The final pipeline is heavy: listener inputs take about 5 minutes per clip on an H200, a picture about 20s per try (up to five tries), and the gate and scene check call a 27B model many times. It is an offline tool, not a live captioning system.
- **Vocabulary.** The 527 AudioSet classes and the 215 drawable families have no word for some sounds (“phone alert”); the scorer’s single-parent ontology file ignores 38 multi-parent labels.
- **Licences.** Most clips are for research use and are not redistributed; Audio Flamingo Next has a non-commercial licence.

9.2 Future work

In order of expected value:

1. **A study with DHH viewers**, as designed in `docs/history/plans/beta_specification.md`: 5–10 viewers, one question per clip (“helped / hurt / neither”), and an ordinal regression of the answer on misses and wrong pictures, which measures β instead of assuming it.
2. **A second annotator** for 30 clips (the packet exists), with κ on the needed decision; below 0.4 the definition itself would need work.
3. **The presence-versus-production question**: test a larger VLM (Qwen3-VL-235B) on exactly this question; a production veto read as a logit margin on the named phrase; a temporal contrast (the onset frame against a control window); an external grounding box with a crop.
4. **A detector that hears under speech and music**, trained on mixtures, tested on the unread fresh set of 422 AudioSet-Strong clips.
5. **Benchmark v2**: at least 300 clips, at least two annotators, the two tagger-guide rules (2-s return, visibility split) written into the guideline, and a sealed test split with a submission scorer.
6. **Display**: a picture with a word, and a “tentative” style for low-confidence pictures, both of which need viewers to evaluate.

10 Reproduction

10.1 Code state

Check out the tag `detector-frozen-2026-10-02`. `config.use_shipped()` sets the final system and `config.use_scored()` the September baseline. The scoring code names variants by short internal codes; the final system is the variant `SHIP8+MD3+WW5+SL` in `benchmark/gold/round13_dev.py`, and this code is used in the commands below (Appendix D explains it). The cluster copies used for scoring are not git repositories; the frozen code was synced from the tag.

10.2 Environment and models

- Conda environment `msproj`: PyTorch 2.5.1, transformers 5.16.1 (Qwen3.x VLMs need ≥ 5.8), diffusers, faster-whisper, panns-inference, easyocr, sentence-transformers. FineLAP needs transformers 4.51.3 and runs in a separate `~/venv_flap` called as a subprocess. The ComfyUI environment is recorded in `docs/envs/comfy.yml`. The `requirements.txt` file predates the final stack and is incomplete.
- FlexSED and DASM run from their own repositories (`~/FlexSED`, `Transformer4SED`), imported in process.

- Model identifiers (Hugging Face): Qwen/Qwen3.8-27B; Qwen/Qwen3-Omni-30B-A3B-Instruct; nvidia/audio-flamingo-next-hf; AndreasXi/FineLAP (revision b419aa2); CPF2/detect_any_sound; Qwen/Qwen-Image-2512; google/owlv2-base-patch16-ensemble; BEATs iter3+AS2M cpt2; PANNs CNN14. Revisions other than FineLAP are not pinned; the Hugging Face cache was used offline.
- Hardware: Bar-Ilan SLURM cluster, partitions with H200 and A100 GPUs. Qwen-Image needs about 57 GB in bf16; the picture check needs the generator and the VLM on one H200 or two cards.

10.3 Running one video

```
python main.py --input clip.mp4 --device cuda
```

`main.py` applies `use_shipped()`. Without `-listener-split`, `src/listener_prep.py` first builds the per-clip inputs the detector needs (FlexSED scores, a first pass with the old pipeline, BEATs and PANNs scores, the listener pools and answers, DASM and FineLAP scores) – about 5 minutes on an H200 – and then runs `pipeline.run`. The output is `data/output/<stem>_augmented.mp4`.

10.4 Running a folder of clips on the cluster

```
sbatch slurm/run_best.sh NAME /path/to/clips      # stages 1-5
python scripts/best_to_protocol.py --name NAME --arm SHIP8+MD3+WW5+SL
python scripts/repaint_shipped.py --tag NAMEbest_v33 --verify
python scripts/recomposite.py --root data/work/shipped_v_NAMEbest_v33 \
    --shipped --require-local-images --modes clean --out OUT
```

The batch runner stops at the list of pictures; the three scripts draw and compose them.

10.5 Reproducing the scores

```
python benchmark/gold/round13_dev.py stage4      # then stage5, score
python benchmark/gold/merged_dev.py
python benchmark/gold/test_vs_ship8.py
python benchmark/gold/parity_check.py SHIP8+MD3+WW5+SL
python benchmark/gold/inspector_trail_export.py \
    --expect DEV=29/58/15/6/7/2/2.056 --expect TEST=24/65/24/4/15/5/2.409
```

The last command re-runs the final system with logging on every decision and fails unless the counts match exactly; it passed on 2 Oct (0 of 159 clips differ). To compute the missing test of the final system against the September baseline, run `benchmark/gold/final_test.py (score)` with the final variant on the cluster, where both per-clip rows exist.

10.6 The Decision Inspector

`docs/inspector2/` is a static page over `data.js` (12.8 MB, 159 clips) that shows, for every gold sound and every candidate of the final system, each step it passed or failed, the value, the bar and the model answers, next to the rendered video (`docs/inspector/media/bysig/`, 350 MB, not in git). The logging hooks (`src/trail.py`) only record and never change a decision. Known gaps: 86 of 156 gate decisions reuse stored verdicts without the raw answers; the depict-event look-alike replies are not stored.

10.7 Where the records are

file	content
docs/history/preregistrations/prereg_round13_detector_push.md	every pre-registration and result, rounds 13–66
docs/history/preregistrations/prereg_v4.md, docs/prereg_round*.md	v4 amendments 1–25, rounds 4–12
docs/history/review/detector_final_2026-10-02.md	plain-language summary of the final system and the last day
docs/history/review/improvements_log_2026-09-30.md	ship chain and rejected ideas, 30 Sept – 1 Oct
docs/history/review/panel*_2026-10-01/, docs/panel_*.md	expert panels
benchmark/gold/annotations/gold_AG.json	the gold labels
benchmark/gold/ledger_*.json	per-item misses and wrong pictures
benchmark/gold/ceiling_ship7.md, visible_weight_sweep.md	ceiling and sensitivity analyses
docs/history/daily_notes/LEDGER_2026-09-26.md	audit ledger of the September phase

11 Conclusion

A deaf viewer of a video misses the sounds whose sources are not on screen, and no existing channel supplies them automatically. This project built a pipeline that does – detection by two detectors checked by two listeners and a witness model, an on-screen decision from the frames, a guarded picture beside the video – from open models with no training; a per-sound benchmark of 159 scored clips; and a metric that prices what a viewer loses. The frozen system finds half of the needed sounds on the development set and 37 % on the test set, with two thirds and one half of its pictures right, and is far cheaper than both showing nothing and the September pipeline. The remaining errors sit where today’s open models reach their limit: sounds no model hears under speech and music, and a vision model that takes “the object is there” for “the object is making the sound”.

Answers to the research questions.

RQ1 – which audio information matters most?

Discrete, sourced events whose source is off screen: sirens, alarms, bells, a baby crying, glass, a knock. The benchmark’s importance rule (an event one can say in a sentence, or danger) captures them; steady place noise (wind, traffic hum) is excluded by design, and it is also where detectors, gate and labels disagree most. Whether DHH viewers rank sounds the same way is untested.

RQ2 – what granularity?

Decide and score at the level of the AudioSet *family* ($\text{depth} \geq 1$), draw the specific source. Family matching was the one de-duplication and scoring rule that generalised without a threshold; the picture names the maker (a barking dog, not “bark”) from a hand table, which beat generic VLM wording 7/7 to 3/7.

RQ3 – can generated augmentations improve accessibility beyond subtitles?

Not shown. What is shown is that the system adds information no subtitle carries (off-screen sounds) at a modelled cost below showing nothing, and that its pictures are recognised at a glance about half the time. A study with viewers is the necessary next step.

RQ4 – how to evaluate?

Per sound, against human labels, with the error split by cause and a cost that prices a

miss against a wrong picture, reported over a range of weights; decisions pre-registered on a development set. Automatic protocols with model-written references or model judges failed: they are circular, decide visibility at chance, or cannot tell a picture from a text tag.

RQ5 – comparison with audio-to-visual approaches?

Audio-to-image models draw the whole scene and cannot leave a visible object out, so they were represented by a blind baseline with the same detector and generator. Against it, the gate removes about half a wrong picture per clip at a small cost in recall (Holm-significant on the September test table); F1 does not move because the gate’s value is in what it does not show.

A Prompts

All questions are asked with greedy decoding and, for Qwen3.8-27B, thinking off. $\langle \text{label} \rangle$ is the family name, lower-cased and cut before a comma or bracket. Source: `src/stage5_cross_modal_analysis/reason.py`, `benchmark/listener_round.py`, `benchmark/gold/listener_variants.py`, `src/stage6_visual_augmentation/group.py`, `depict.py`, `verify.py`, `gen_screen.py`.

Gate: name vote

“These frames are from the moment a sound of $\langle \text{label} \rangle$ was heard. Name the thing in these frames that is making that sound. Answer with a short noun phrase of at most 5 words. If nothing that could make that sound is visible in these frames, answer exactly: nothing.”

Gate: follow-up (text only)

“Sound heard: $\langle \text{label} \rangle$ / Thing visible in the video: $\langle \text{named} \rangle$ / Could that thing be what is making that sound? Answer yes or no.”

Gate: a/b vote (asked in both orders)

“These frames are from the moment a sound of $\langle \text{label} \rangle$ was heard. Judge from the frames alone. (a) you can SEE $\langle \text{label} \rangle$ happening on screen – the source is in the frames and visibly making that sound (b) $\langle \text{label} \rangle$ is not visibly happening in these frames. Answer with the letter only.”

Gate: describe vote

“Describe what is happening in these frames in one sentence. Name what you see and what it is doing. No more than 25 words.”

Speech context (text only, both orders)

“... Are they (a) reacting to that sound or talking about it, or (b) not referring to that sound?”

Scene check (witness rule; logit read)

“Could the sound of $\langle \text{label} \rangle$ plausibly be heard in this scene? Answer yes or no.” and the twin

“Could the sound of $\langle \text{label} \rangle$ NOT plausibly be heard in this scene? Answer yes or no.”

Listener yes/no (Qwen3-Omni, AF Next)

“Is the sound of $\langle \text{family} \rangle$ present in this recording? Answer yes or no.”

Listener inventory

“List every distinct non-speech sound you hear in this recording, one per line, most prominent first.” (64 new tokens)

Grouping (Qwen3-Omni, both orders)

“You hear a $\langle \text{lab} \rangle$ sound near the start and again near the end of this recording. Is the $\langle \text{lab} \rangle$ near the end $\langle o_1 \rangle$, or $\langle o_2 \rangle$? Answer with exactly one word: $\langle w_1 \rangle$ or $\langle w_2 \rangle$.” with the options “the same continuing sound as at the start (for example one alarm or engine with a short pause)” and “a new, separate event (for example a second bark, knock or shot)”.

Depict-event

“Is " $\langle \text{event} \rangle$ " visibly happening in these frames? Answer yes or no.” and “Do these frames show no " $\langle \text{event} \rangle$ " happening? ...”; then (text only) “Could the sound of $\langle B \rangle$ be mistaken for the sound of $\langle A \rangle$?” in both directions.

Picture check

“Look at this picture. What is the main thing shown in it? Choose exactly one option. ... Answer with the letter only.”

Picture prompt tail

“, the whole thing fully in frame, caught at the moment it makes the sound with its visible effect, one large subject filling the picture, plain white background”; negative prompt: scenery, landscape, background scene, room interior, street, buildings, sky background, text, letters, watermark.

B Configuration of the final system

Values set by `config.use_shipped()` (`config.py`, lines 503–572), read by running it. “ $\rightarrow$ ” gives the module default and the shipped value.

variable	value	meaning
AED_THRESHOLD	0.175	BEATs extraction bar
AED_MIN_DUR	0.5 $\rightarrow$ 0.3	shortest span (s)
DISPLAY_THRESHOLD	0.35	display bar (stage 5)
FLEXSED_BAR	0.8	FlexSED extraction bar
FLEXSED_VETO	0 $\rightarrow$ 0.3	FlexSED clip veto
PANNS_VETO	0.05	PANNS clip veto on FlexSED-only spans
TWIN_MAX	True	twin confidence = stronger side
MIRROR_VETO	0.7	mirror veto bar
LISTENER_CONFIRMED_MIRROR	True	listener can undo the mirror veto
KEEP_NEEDS_V4	True	a keep must also be named in an inventory
MASKED_WEAK_VETO, _AF	True	masked-weak veto
DASM_CLIP_VETO	0.0839	DASM clip veto
DASM_LOCAL_VETO	0.35	witness rule bar
DASM_LOCAL_KEEP	both	both listeners can witness
SCENE_FIT_LOGIT	True	scene check by logit margin
KEEP_NEEDS_V4_ALL	onto	inventory keep incl. descendants
KEEP_NEEDS_V4_ALL_DASM_KEEP	True	DASM can keep it
BAND_TWIN_PULL	0.5	band-twin pull
CONTINUATION_VETO	0.5	continuation veto
LISTENER_RESCUE	True	band rescue
LISTENER_LO	0.4 $\rightarrow$ 0.5	lowest FlexSED peak rescued
LISTENER_RULE	TIER	Qwen alone ≥ 0.6 , both below
DASM_RESCUE, _NEW_ONLY	True	DASM rescue of new families
FINE LAP_VETO	0.329	FineLAP veto on rescued spans
LISTENER_DASM_VOTE / _BAR	True / 0.575	DASM vote on rescued spans
LISTENER_ONCE	True	one rescue per family per clip
ONSET_CAM, ONSET_MONOTONE	True	occlusion onsets, never earlier
MAX_SPAN	8.0 $\rightarrow$ None	no length cap
LABEL_FILTER	depictable	what may be drawn
VLM_MODEL	Qwen/Qwen3.8-27B	gate, subject, checks
VLM_THINKING	False	thinking off

variable	value	meaning
VISIBILITY_STRETCH / _RULE	5.0 / majority	gate stretches and vote
KINSHIP_DIRECTED	True (scored: False)	kinship direction
VIDEO_BACKEND	owlv2	stage-2 objects
DEPICT_EVENT	True	depict-event drop
MERGE_GAP	2.0 $\rightarrow$ 2.5	repeat merge (s)
MIN_DWELL	1.5	minimum on screen (s)
MAX_AFTER_END	1.0 (scored: None)	cap past the real end (s)
GROUP_ASK / GROUP_MAX_GAP	True / 8.0	grouping by Qwen3-Omni
MAX_SLOTS	3	simultaneous pictures
GEN_MODEL	Qwen/Qwen-Image-2512	picture model (FLUX retired)
RESOLUTION	1024 $\times$ 1024	picture size
PICTURE_V3, _SCENE, _FINAL, _MAKER	True	subject writing
PICTURE_VERIFY / _TRIES	True / 5	check and redraw
CONFIDENCE_FADE	False	full opacity

C Sources of the numbers

Every number in the main body comes from one of the files below (paths relative to the repository root). Counts of the final system are also reproduced exactly by the decision-trail export (`docs/inspector2/data_parity.json, "exact": true`).

numbers	file
Table 1 (set sizes, needed sounds)	<code>benchmark/gold/annotations/gold_AG.json</code> , counted with <code>benchmark/gold/score_per_sound.py</code>
labelling rules (Section 3.1)	<code>benchmark/gold/tool_template.html</code> , <code>tagger/index.html</code> , <code>docs/history/plans/metric_per_sound.md</code>
scoring rules, cost (Section 3)	<code>benchmark/gold/score_per_sound.py</code>
bootstrap and pass rules	<code>benchmark/gold/dev_candidates_check.py</code> ; <code>benchmark/gold/holm_table.py</code>
final system, Table 4	<code>docs/history/review/detector_final_2026-10-02.md</code> ; <code>docs/inspector2/data_parity.json</code>
September baseline, Table 4	<code>docs/history/review/improvements_log_2026-09-30.md</code> ; <code>benchmark/gold/final_test_ship8.json</code>
intermediate-version p (Section 6.1)	<code>benchmark/gold/final_test_ship8.md</code>
on-screen weight (Section 6.2)	derived from Table 4; earlier versions in <code>benchmark/gold/visible_weight_sweep.md</code>
Table 5 (fate of misses)	<code>docs/history/review/inspector_trail_ready.md</code> ; <code>docs/inspector2/data.js</code>
Table 6 (ceiling)	<code>benchmark/gold/ceiling_ship7.md</code>
Table 7 (gate value)	<code>benchmark/gold/holm_test_final_v33_test_bench.json</code>
gate accuracy, failure modes	<code>docs/history/review/vlm_panel_2026-10-01/f1_audit.md</code>
listener agreement on the screening set	<code>docs/history/review/improvements_log_2026-09-30.md</code> ; <code>docs/history/preregistrations/prereg_round13_detector_push.md</code>
onset timing	<code>benchmark/eval_dcase_onset.json</code>

numbers	file
DCASE visibility check	benchmark/eval_dcase_visibility.json
picture glance test	docs/history/preregistrations/picture_v3_prereg.md
picture check validation	docs/history/preregistrations/freeze_picture_setup_2026-09-25.md
label re-checks, missing-sound checks	docs/history/preregistrations/prereg_round13_detector_push.md
test-set reads (Section 8)	docs/history/preregistrations/prereg_round13_detector_push.md; docs/history/earlier_drafts/thesis/ch5_results.md
configuration (Appendix B)	config.py (use_shipped())
development history (Appendix E)	docs/history/ (preregistrations, panels, daily notes, review)

D Internal names used in the development record

The development logs in `docs/history/` and the scoring code use short codes. The main body of this report does not; this table maps each code to the plain name used in the report.

code	plain name / meaning
<i>Systems and versions</i>	
D', SHIP8+MD3+WW5+SL	the final system (<code>config.use_shipped()</code> , tag <code>detector-frozen-2026-10-02</code>)
D	the final system with the scene check read as text
B, SHIP8+MD3	the version before the last three rules (witness rule, scene check, logit reading)
B0, B0r	the September baseline (<code>config.use_scored()</code>); B0r = B0 re-run in the same job as a candidate
B1	the September baseline with a BEATs self-veto (rejected)
SHIP1–SHIP8	successive accepted versions (Appendix E)
v1, v2, v3, v4	earlier versions of the system (August–September; Appendix E)
<i>Data</i>	
DEV-49, TEST-60	the older parts of the development and test sets
DEV2, TEST2	the newer parts (tagger clips)
merged DEV / TEST	the development set (71 clips) / the test set (88 clips)
the 415, held-out slice B	the screening set (415 AudioSet-Strong clips) 30 AudioSet-Strong clips with a masked sound, reported separately
<i>Components</i>	
TO1+F7F8, TIER	listener rescue of weak FlexSED detections (Qwen alone for strong runs, both listeners for weak)
V4	a listener’s open list of heard sounds (“inventory”)
N2b	masked-weak veto
DR2	DASM rescue of new families
K-V4	a listener keep must also be named in an inventory
DV	DASM clip veto
BTP	band-twin pull (start a late picture at an earlier weak run)
CONT	continuation veto
FLAP	FineLAP veto on rescued sounds
K4AO, K4A-D	inventory keep (drop a picture neither listener lists; K4A-D: unless DASM hears it)

code	plain name / meaning
MD3	shortest detected sound 0.5 $\rightarrow$ 0.3s
GRP, GROUP	same-family grouping by Qwen3-Omni
WW, WW5	witness rule; WW5 = witness rule with the scene check
SL, 60L	scene check read from the yes/no logits
DEPICT	depict-event drop
EXPECT	“expected sound” rules: name sounds expected off screen, then confirm (rejected)
<i>Process</i>	
Round n , Rn	a numbered, pre-registered development round
amendment n	a numbered change to the September pre-registration (<code>prereg_v4.md</code>)

E Development history

This appendix is the record of how the final system was reached: how the data grew, the working rules, the earlier versions of the system, every accepted change, the main rejected ideas, the history of the pictures and of the evaluation, and a round-by-round table. Internal codes from the development logs appear here in parentheses so that each row can be found in `docs/history/`; Appendix D maps them to plain names.

E.1 How the data grew

Clip-level benchmark (August–September). 274 short clips (10–20s) were labelled by the author into four scenarios: unseen ambient (48 clips), mixed (25), seen ambient (132) and no ambient (69). Four sourcing strategies (keyword scraping, AudioSet strong segments, curated film scenes, UnAV-100 overlap moments) and an automatic filter all landed at 15–20 % usable clips, which is why the set stopped at 274 rather than the planned 300. A test set of 100 clips (25 per scenario) was used by the early systems and the model judge (Appendix E.7).

Per-sound labels (22 September). After the judge-based evaluation was abandoned, clips were labelled per sound. The first export had 139 clips: 109 benchmark clips and 30 AudioSet-Strong clips with a consequential sound at least half under speech or music (“slice B”, always reported separately). The 109 were split by provenance into 49 development clips (“DEV-49”, clips already touched by earlier runs) and 60 test clips (“TEST-60”, labelled after the system was frozen). The September test table (Section 6.5) is on TEST-60.

New clips (29 September – 1 October). 50 more clips were labelled with a stand-alone tool in three batches: 31 YouTube clips found by scene type and 19 AudioSet-Strong evaluation clips. They were split into 22 development (“DEV2”) and 28 test (“TEST2”) clips by a hash of the clip name or by balanced pairs, and merged on 30 Sept into the development set ($49 + 22 = 71$ clips) and the test set ($60 + 28 = 88$ clips) used in the body.

Table 11: Parts of the two evaluation sets.

part	clips	sounds	needed	imp. 3	clips w. needed	unseen/mixed/seen/none
DEV-49	49	95	38	14	24	16/9/16/8
DEV2	22	59	20	3	14	8/7/7/0
TEST-60	60	94	43	11	28	13/15/12/20
TEST2	28	60	22	2	15	8/8/10/2

Needed-sound count. The development count moved during the last week: 55 at the first merged scoring, 58 after the visibility re-check (three sounds changed from seen to needed), 57 after a control round, and back to 58 after a correction. Every system was re-scored on each version; the test labels were never changed.

AudioSet sets. A calibration set of 280 AudioSet-Strong clips decided detector rounds 4–12 with a cost on timed AudioSet labels. Round 12 showed that under a corrected version of that cost the detector of the time was worse than showing nothing (1.650 vs 0.986 per clip), because AudioSet labels only a few sounds per clip; the set was retired as a decision test. The screening set of 415 clips was then used only for single audio rules, and a fresh set of 422 clips built for a last confirmation was never scored.

E.2 Working rules

From 13 Sept the following rules governed every change (`docs/history/plans/plan_robustness.md`, `docs/history/preregistrations/prereg_round13_detector_push.md`):

1. **Pre-register.** Before a run, write the idea, the exact rule, the data it is decided on and the pass bar, and commit it. Results are appended below the pre-registration, never edited into it.
2. **Decide on development data only.** The development set (or, for audio-only rules, the held-out 415) decides. The test set is read after a change ships and is only reported. The author had to restate this rule on 1 Oct after a test-informed choice (Section 8).
3. **General rules only.** A rule must make sense for any video, run on one video at a time, and not name a clip or a single sound class (a rule that only ever fires on bells was rejected for this reason, Round 64).
4. **Compare with the shipped pipeline.** Every candidate is scored as a full-pipeline variant on the same clips as the current shipped system, not on a proxy; a component screen may stop an idea early but cannot ship it.
5. **Hits first.** Losing a few hits is acceptable only if many wrong pictures go (the fewer-pictures clause, Section 3.5).
6. **Consult and record.** Major choices were reviewed by panels of independent model reviewers (Claude instances with assigned roles, two to five per panel, one to four rounds), whose reports are in `docs/history/review/` and `docs/panel_*.md`; every decision and its reason is in a dated log.

E.3 Phases

Table 12: Project phases. Scores in each row are in that phase’s own metric and are not comparable across rows.

dates	system	scored by	outcome and why it changed
06 Aug	proposal: 7 stages (audio, VLM, ASR, BEATs/PANNs, reasoning, FLUX/SDXL, evaluation)	planned: VLM describes, LLM writes reference, LLM judges	pivot from an earlier speech-illustration project

dates	system	scored by	outcome and why it changed
07 Aug	v1: PANNs (bar 0.15) → Openverse image retrieval → side panel; no gate	demo clips	needed a “not seen” decision
07 Aug–07 Sep	CLIP zero-shot visibility gate; clip-level benchmark (274 clips)	clip-level gate accuracy	CLIP gate 49.3%: 71 of 103 seen clips still augmented – “object presence is not event visibility”
08–11 Sep	PANNs + OWLv2 + per-sound VLM gate (Qwen2.5-VL-7B, “name it, don’t confirm”); SDXL	100 clips × 3 systems, Mistral-7B judge 0–4, model-written reference	blind 2.94 > caption 2.68 > gated 2.41; retrieval beat SDXL by +0.17/+0.18; reference later found circular
12–17 Sep	v3: BEATs (replaced PANNs after 11 of 14 complaints traced to its labels), hysteresis + occlusion onsets, 7B gate with 3 votes, FLUX.1-schnell	human-grounded judge reference, dev 174 / test 100	tie 3.17 vs 3.16; gate wins where nothing is due (+0.26), loses where a picture is due (−0.24)
18–21 Sep	v4: one state-of-the-art swap per stage; Qwen3.8-27B gate; FLAM and PretrainedSED detectors tried; rubric-enforced judge	judge with a cap on redundant pictures	PretrainedSED rejected three times; 13 silent bugs found (amendment 3), all earlier judge rows kept as history only
22–23 Sep	per-sound gold (139 clips); BEATs ∪ FlexSED (bar 0.8), FlexSED veto 0.3, PANNs veto 0.05	per-sound F1, then viewer cost (amendment 9)	F1 difference null (+0.028); wrong pictures and precision significant
24–27 Sep	monotone onsets, no length cap; final September test table (amendment 21); Qwen-Image frozen as picture setup	Holm-corrected table on TEST-60	“Holm wins, F1 null” (Section 6.5)
27 Sep	detector rounds 1–3: cascade, PE-A-Frame, Qwen3-Omni as verifier	DEV-49 + AudioSet 280 / 415	none passed; detection declared closed, reopened next day
28 Sep	rounds 4–12 on AudioSet sets: EAT, Dasheng, DASM, SAM-Audio, 17 ideas, local contrast, DASM rescue, masker tests	cost on AudioSet labels	all failed on the 415; round 12 showed the AudioSet cost cannot track the task: harness retired

dates	system	scored by	outcome and why it changed
28–30 Sep	rounds 13–14: listener rescue (Qwen3-Omni, Audio Flamingo Next) + DASM vote; tagger clips added	DEV-49, then merged DEV	first gains in hits (TO1+F7F8)
30 Sep–01 Oct	ship chain SHIP1–SHIP8 on merged DEV/TEST	merged DEV, pass rules	DEV 18/51/3.690 → 28/21/2.282 (Table 13)
01–02 Oct	rounds 46–66: display rules, witness rule, scene check, many gate ideas	merged DEV, held-out 415	the final system: DEV 29/15/2.056; frozen 2 Oct 01:20 UTC

E.4 The shipped chain

Table 13 is the backbone of the history: every change that entered the final system, in order, with its development and test numbers on the merged sets. Development numbers use the corrected gold (58 needed sounds); test numbers are the reads made after each change shipped. Sources: `docs/history/review/improvements_log_2026-09-30.md`, `docs/history/review/detector_final_2026-10-02.md`, `docs/history/preregistrations/prereg_round13_detector_push.md`.

Table 13: Changes that shipped, from the old pipeline to the final system. Hits / wrong / cost; DEV hits out of 58, TEST out of 65. The p in the TEST column is one-sided, against the old pipeline B0r for SHIP5–SHIP8 and against B for D and the final system.

step (name in logs)	what it adds	merged DEV	merged TEST
old pipeline (B0r)	September test-table system: BEATs $\cup$ FlexSED, FlexSED and PANNs vetoes, Qwen3.8 gate	18 / 51 / 3.690	21 / 40 / 2.909
TO1+F7F8 (R14)	listener rescue of weak FlexSED runs (tiered: Qwen alone above 0.6, both listeners below), one rescue per family, mirror veto with listener keep, DASM vote on rescues	26 / 45 / 3.070	22 / 38 / 2.818
+ N2b, DR2, K-V4 (R16–21)	masked-weak veto; DASM rescue of new families; a listener keep must also be named in an inventory	27 / 38 / 2.817	23 / 36 / 2.727
+ DV (R28)	drop a sound DASM never hears in the clip	27 / 31 / 2.620	23 / 34 / 2.682
+ BTP (R30)	start a late picture at an earlier weak run of the same sound	28 / 30 / 2.535	23 / 33 / 2.659 (p 0.033)
+ CONT (R31)	drop a picture that is only a later piece of a sound already going on	28 / 25 / 2.394	24 / 32 / 2.591 (p 0.025)
+ FineLAP veto (R31)	a 2025 frame-level model double-checks rescued sounds	28 / 24 / 2.366	24 / 31 / 2.568 (p 0.015)
+ K4A-D = SHIP8 (R35)	drop a picture neither listener names, unless DASM clearly hears it	28 / 21 / 2.282	23 / 29 / 2.568 (p 0.017)
+ merge gap 2.5 s	continuing sounds with short breaks no longer drew a second picture	28 / 20 / 2.254	23 / 28 / 2.545
+ grouping ≤ 8 s (R47)	Qwen3-Omni: same continuing sound or a new event?	28 / 18 / 2.197	22 / 26 / 2.545
+ shortest sound 0.3 s (R48)	short real sounds (a laugh burst) were thrown away	29 / 18 / 2.141	22 / 26 / 2.545

step	what it adds	merged DEV	merged TEST
+ depict-event (R57)	drop a picture whose event is visibly happening	29 / 17 $\rightarrow$ 18 ^a / 2.141	22 / 26 / 2.545
= B (SHIP8+MD3)	base of the last rounds	29 / 18 / 2.141	22 / 26 / 2.545
+ witness rule + scene check (R53+R60) = D	weak sounds need DASM, both listeners, or one listener and a plausible scene	29 / 14 / 2.028 ^b	24 / 23 / 2.386
+ logit readout (R60L) = final system	scene check read as a probability, no cut-off answers	29 / 15 / 2.056	24 / 24 / 2.409 (<i>p</i> 0.034 vs B)

^a Round 57 scored 29/17 with 26 listener answers missing from a cache; after the cache was filled (1 Oct evening) the same system scores 29/18/2.141, so its net development effect is zero. ^b D’s 2.028 contains one gain caused by a cut-off answer (“based on the”) that removed a wrong *Glass* picture; the logit readout of the final system cannot be cut off, so that picture returns (+1 wrong). The pass bar of Round 60L was set against B, not against D, for this reason (prereg lines 4504–4542).

What the table shows. From the old pipeline to the final system the development set gained 11 hits and lost 36 wrong pictures (cost 3.690 $\rightarrow$ 2.056). The test set gained 3 hits and lost 16 wrong pictures (2.909 $\rightarrow$ 2.409). About a third of the development gain in cost carried over to the test set; with well over a hundred development variants tried, some of the development gain is expected to be selection luck (“winner’s curse”), and the test numbers are the better estimate of what a new video would see.

E.5 What was tried and rejected

Most ideas failed. Table 14 groups the main ones by the question they tried to answer, with the reason from their own pre-registered test. All were decided on the development set or the held-out 415.

Table 14: Main rejected ideas (selection). Hits / wrong / cost on merged DEV unless stated.

idea (round)	what it did	why it failed
<i>Hear more sounds</i>		
other detectors (Sep, rounds 1–8)	FLAM, PretrainedSED (5 backbones), AST, CED, SSLAM, EAT, Dasheng, PE-A-Frame, CLAP verifier, tagger ensembles	recall rose only with many more false spans; PretrainedSED lost three times; ensembles mis-scaled
source separation (Sep, R7)	Demucs, SAM-Audio in front of BEATs	separated audio is out of the detectors’ training distribution: recall under speech fell 9.5 % $\rightarrow$ 4.8 %; SAM-Audio failed its quality check (music never removed)
listener as verifier (R13)	ask Qwen3-Omni yes/no, multiple choice, open list on every weak candidate	finds 5–8 dropped real sounds but adds about 3 wrong pictures per hit
newer listeners (R31–34)	MOSS-Audio, SpotSound, Kimi, Step-Audio	3–10 wrong per extra hit; SpotSound said “Yes” to everything
EXPECT chain (R40–42b)	the scene VLM or a listener names sounds expected off screen; FineLAP times them; DASM confirms	best DEV 31/26/2.254 (+3 hits), but TEST 25/35/2.614 (worse); the gate cannot refuse a wrong off-screen name

idea	what it did	why it failed
short twins, band lists (R49, R56)	add weak or short detections both models half-hear	BANDLIST 0 hits, +6 wrong; TWIN-SHORT 27/36/2.761
tagger ensemble (R63)	calibrated EAT/SSLAM ensemble on BEATs windows	on the 415: 33 % fewer spans but recall 0.152 vs 0.197
<i>Remove wrong pictures</i>		
stricter witnesses (R53 variants)	require DASM or both listeners for every weak sound	27/13/2.113: −5 wrong but 2 real hits lost (fails the 3:1 clause)
named veto, relabel gate (R39–41)	drop or rename a picture when the gate or the listeners name a different sound	real hits lost for no cost gain
held-out family prior (R41)	drop families a detector gets wrong > 70 % of the time on the 415	every affected picture had been heard by a listener: no change
<i>Better on-screen decision</i>		
other gate signals (Sep, am. 14–20)	OWLv2 or SAM 3 vote, video input, synchrony, “any stretch” rule	each removed exactly as many needed sounds as wrong pictures it saved (break-even at $\beta = 2$)
SSL-SaN, SigLIP, BOX (R30–38)	audio-visual localiser; picture-vs-frame similarity; VLM draws a box on the source and checks the crop	each un-silenced real visible sounds; BOX-2 in the full pipeline 29/26/2.366
synchrony (R38)	Synchformer: does on-screen motion move with the sound?	as a veto, +3 needed kept but 5 visible calls lost; as an “add seen” vote, on the bar but changed no shipped picture
human-style chain (HUMAN, R50, R66)	name the source, find it, zoom, ask “is it making the sound now?”	the VLM names “nothing” on most stretches; NAME-ALL passed its gate screen but lost 3 real hits in the pipeline (26/12/2.141)
sign/event and logit gate (R54, R62)	ask for visible effects; read the gate as a logit margin	letter-position bias or “no” habit; best AUROC 0.647, below the 0.65 bar
look-alike reasoning (R61)	“no train here, so it is the washing machine”	works on the laundromat, but also explains away real sounds (a crying sound “explained” by a visible parrot)
bell rule (R64)	concealed actions (a bell in a tower) count as not visible	passed DEV (30/14/1.972) but fires only on bells: not general, not shipped
<i>Timing and grouping</i>		
change-point onsets (R33)	SEBB onset post-processing	7 hits lost
earliest onset (R38), repeat rules (R31, R35)	start at the first detector to hear a sound; require silence before a repeat	hits lost or no net gain
pause questions (R51, R55)	ask whether a 2-s gap is a pause or a new sound	neither the listener nor the detectors can tell a pause from a dropout
context ± 5 s (R59)	give the listener more context	it names what it <i>sees</i> , not what it hears

The main lesson. Two patterns recur in this table. First, the missing sounds can often be found – a listener asked about the whole clip names the hammer, the train, the bell – but every way of adding them brings two to four wrong pictures per hit, because nothing downstream can tell a right off-screen name from a wrong one. Second, every change to the gate moves along the same trade-off line: it silences more on-screen sounds only by silencing about as many needed ones. Section 7 measures both.

E.6 Pictures

The picture model changed four times: Openverse retrieval (v1), SDXL and SDXL-Turbo, FLUX.1-schnell (chosen over PixArt- Σ on twelve real prompts, 13 Sept), and Qwen-Image-2512 (frozen 25 Sept, shipped 28 Sept). On 9 Sept a retrieval-versus-SDXL ablation on the same 100 clips favoured retrieval by +0.17 (gated) and +0.18 (blind) judge points, but 55 % of retrieved images carried a non-commercial or no-derivatives licence and 595 uses drew on only 60 distinct images, so generation was kept. The decisive picture test was a blind *glance test* by the author on 25 Sept: 54 sounds from 50 never-annotated clips, each picture shown for 1.5 s at 384 px with no name, and the author typed what the sound was. FLUX was recognised 14/54 times; Qwen-Image with the same text 26/54 (+0.222 [+0.074, +0.370]); Qwen-Image with a richer subject text 32/54, but that text invented an object (a thud drawn as a door) and was rejected. One of the 26 Qwen-Image pictures was later found to send a false message (a telephone bell drawn as a desk bell), so the +22-point result is reported as “not clean”. A planned sealed confirmation sitting was cancelled on 28 Sept and replaced by the check-and-redraw loop of Section 5.6. A wording test the same day kept the hand-written subject table: 7 of 7 known cases right, against 3 of 7 for two generic VLM wording methods (docs/history/analyses/picture_sense_test_2026-09-28.md).

E.7 Earlier evaluation by model judge (superseded)

The proposal planned an automatic evaluation: a VLM describes the augmented video, an LLM writes a reference of what the viewer should learn, and an independent LLM scores the description against the reference on a 0–4 scale. It was built and run, and it failed in instructive ways:

1. **A reference written by the system’s own models is circular.** The first reference was written by the system’s VLM from the system’s detections; it rewarded whichever system repeated the detector (blind 2.94 > caption 2.68 > gated 2.41 on 9 Sept). A second judge of another family agreed with the first (quadratic $\kappa = 0.753$), which shows only that two judges can share a bias.
2. **A reference written by independent models cannot decide visibility.** A reference built by an audio LLM, CLAP, a different VLM and a different LLM agreed with the human “nothing is missing” label on 55 of 100 clips – chance. An audio-visual model asked with and without the soundtrack reached 49.5 % balanced accuracy (bar 67.7 %); a “gap-closing” score had AUROC 0.529 (bar 0.75).
3. **The rubric was asymmetric.** A withheld picture scored 0 where one was due, a redundant one about 3 where none was due, so the rubric punished staying silent – the very behaviour the gate exists for. A capped rubric (19 Sept) and a human-grounded reference fixed the direction but not the noise.
4. **Judges cannot tell a picture from a text tag.** The best judge (Gemma-4-31B seeing the pictures directly, passing all three trust checks against the author’s labels) scored pictures and text tags alike (−0.04 [−0.22, +0.14]), because it reads a tag as the label. Six automatic picture checkers also failed calibration against the author (for example $\kappa = 0.21$ to 0.45 against bars of 0.5).

A five-reviewer scoring panel on 19 Sept moved the primary score to per-sound matching against human labels (Section 3) and kept the judge as a secondary instrument. The answer to RQ4 is built on this history.

E.8 Comparisons between versions

The final development phase compared each accepted change with the version before it and, early on, with the September baseline (“B0r”). Paired clip bootstrap on the test set, 2000 draws; d = mean per-clip cost difference; one-sided p (a two-sided value is about twice as large).

comparison	d	95 % interval	one-sided p
final system vs the version before its last three rules (B)	-0.136	[-0.295, 0.000]	0.034
same, scene check read as text (D) vs B	-0.159	[-0.341, -0.023]	0.0145
intermediate version (SHIP8) vs September baseline	-0.341	[-0.636, -0.045]	0.017
first listener stack (TO1+F7F8) vs September baseline, pre-registered	-0.091	[-0.250, +0.045]	0.132

Version B scored 29 / 18 / 2.141 (development) and 22 / 26 / 2.545 (test); D scored 29 / 14 / 2.028 and 24 / 23 / 2.386. The four early test reads of 30 Sept were also corrected for multiplicity (Holm); none was significant after correction.

Repeat-merge gap. Re-reading the same saved pictures at different merge gaps moves wrong-picture counts but not the ranking (hits / wrong / cost):

gap (s)	SHIP8, dev.	baseline, dev.	SHIP8, test	baseline, test
1.5	28 / 23 / 2.338	18 / 57 / 3.859	23 / 30 / 2.591	21 / 44 / 3.000
2.0	28 / 21 / 2.282	18 / 51 / 3.690	23 / 29 / 2.568	21 / 40 / 2.909
3.0	28 / 18 / 2.197	17 / 43 / 3.521	22 / 27 / 2.568	20 / 33 / 2.795

The baseline gains more from a longer gap (many of its pictures are repeats), so the final system (gap 2.5 s) against the baseline (gap 2.0 s) is not a like-for-like comparison.

E.9 Round-by-round record

One row per round or main arm, in order. Results are hits / wrong / cost on merged DEV (55 or 58 needed, see Appendix E.1) unless marked; “415” = held-out AudioSet-Strong screen; “old DEV” = DEV-49. Full text, bars and line numbers: docs/history/preregistrations/prereg_v4.md (amendments), docs/prereg_round4..12*.md, docs/history/preregistrations/prereg_round13_detector_push.md (rounds 13–66). Counted from that file: rounds 13–28 ran 129 new full-pipeline variants (106 on DEV-49 alone), rounds 29–41 ran 21 more and 42 component screens, rounds 42–57 ran 5 full variants and about 30 screens, rounds 58–66 four full variants and 15 screens.

round	idea	result	verdict
<i>September amendments (prereg_v4)</i>			
am. 5	per-sound gold, F1 headline	$\Delta F1 +0.028$ [-0.022, +0.074]; precision, false alarms significant	F1 null
am. 8	FlexSED union, bar 0.8	passes 3/3 rules	shipped
am. 9	viewer cost, $\beta = 2$ (post hoc)	gated 3.36 vs blind 4.24; silence 2.79	reported metric
am. 10	FlexSED veto 0.3	4.12 $\rightarrow$ 3.59	shipped
am. 11–15	per-family bars; speech-removed view; family silence; any-stretch rule; OWLv2 vote	each worse or neutral	rejected
am. 16	PANNs veto 0.05	3.02 / 3.10	shipped
am. 17	tagger ensemble	CED mis-scaled	rejected
am. 18–20	visibility signals (video, SAM 3, synchrony)	all at break-even 2:1	rejected
am. 21	final September TEST table	Table 7	reported
am. 22–25	cascade, PE-A-Frame, Qwen3-Omni verifier	17 hits, cost 4.82; AUROC 0.53; AUROC 0.661	rejected
<i>Rounds 4–12 (AudioSet harness, 28 Sept)</i>			
4	BEATs self-veto replaces PANNs veto	415 neutral; later worse on DEV and TEST (2.83 vs 2.63, p 0.043)	reverted 30 Sept
5	EAT, Dasheng	415 +0.087	fail
6	DASM as detector	415 +0.294 [+0.111, +0.511]	fail (kept as witness)
7, 7b	SAM-Audio separation	quality check failed	stop
8	17 ideas (I1–I10)	closest I7 -0.039, p 0.061	fail

round	idea	result	verdict
9	local contrast J2	415 pass, DEV 14→13 hits	not shipped
10	DASM rescue R0–R7	9 of 61 targets reachable	hold
11	masker tests	all worse	no pick
12	cost v2 on AudioSet	shipped 1.650 vs nothing 0.986	harness retired
<i>Rounds 13–28 (29–30 Sept)</i>			
13	listener rescue, many question forms; R13-1 twin-max	~3 wrong per extra hit; R13-1 TEST = B0r	R13-1 kept inside TO1
14	TO1+F7F8	old DEV 18/24/2.45 vs 2.78; merged 23/48/3.155; TEST 22/38/2.818 (p 0.13)	shipped (SHIP1)
16	N2b masked-weak veto	23/44/3.042; TEST 22/35/2.750	shipped
19–20	DR2 DASM rescue	24/44/2.986	shipped (SHIP2)
21	K-V4 inventory keep	24/41/2.901	shipped (SHIP3)
22–27	N2c, stacks, K2 variants	N2c 22/38/2.930 lost a hit	rejected
28	DV DASM clip veto	24/34/2.704; TEST 23/34/2.682	shipped (SHIP4)
<i>Rounds 29–41 (30 Sept – 1 Oct)</i>			
29	F8-U, DV-L, DV-G; V4D; GA	+2 hits for +7 wrong; no-op; –5 hits; STOP; STOP	rejected
30	BTP band-twin pull	25/33/2.620; TEST 23/33/2.659	shipped (SHIP5)
30	K4AO; PIC-SIM; RT	held; nothing flips; no-op	held / rejected
31	CONT continuation veto	25/28/2.479; TEST 24/32/2.591	shipped (SHIP6)
31	FineLAP veto 0.329	25/27/2.451; TEST 24/31/2.568	shipped (SHIP7)
31	RPT-S, PMC, N2c, DVG, PTC, SUBJ, OM, SPOT, H2	screens or arms without gain	rejected
32	FineLAP in DASM's seat; DV2	+11 wrong; Thunder lost	rejected
33	SPOT-V, change-point onsets, K4A-4E, SSL-SaN gate	all “Yes”; –7 hits; no-op; STOP	rejected
34	MOSS-Audio as listener	+4/–9 needed, +63 other	rejected
35	DBX, DBR	premise wrong; cost not lower	rejected
re-check	visibility re-check (gold)	3/14 flips; control 0/20	gold change
35	K4A-D (K4AO + DASM keep)	28/58, 21/2.282; TEST 23/29/2.568 (p 0.017)	shipped (SHIP8)
36–37	DASM impact peaks + VLM naming; BOX gate	+2 cross; –7 seen silenced	rejected
38	CF, BOX-2, GBTP, SHIP8+BOX2, SYNC, SYNC-2	best BOX-2 arm 29/26/2.366	rejected
39	DETACHED-ADD, CONTRAST, MAKER-VIS, RELABEL-GATE	+7 wrong; +142 other; –9 hits; hit lost	rejected
40–40e	EXPECT chain	A4: DEV 31/26/2.254, TEST 25/35/2.614	not shipped
41	AGREE, NAMED-VETO, PRIOR415	below bar; hits lost; no change	rejected
<i>Rounds 42–57 (1 Oct)</i>			
42	held-out check of EXPECT-A4 audio chain	74/92 = 0.80 precise vs 0.38–0.39 raw	diagnostic
42b	EXPECT-A5 (7 vouched families)	DEV 30/21/2.169; TEST 23/31/2.614	not shipped
43–43c, 50	gate gold-set variants (HUMAN, HUMAN-2, HUMAN-ADD)	one sound short of the bar at best	rejected
44	AVNAME (eyes + ears name the sound)	rename 15/34; drop 27/17/2.225, TEST no change	rejected
45	AGREE-EARS (second listener family)	DEV 31/24/2.197; 415 0.833 < 0.85; TEST 2.659	rejected
46, 46b	gold scope (blind rating of wrong pictures)	≈1 in 6 is a real unlisted sound	no gold edit
–	merge gap 2.5 s	28/20/2.254	shipped
47	GRP-A grouping by Qwen3-Omni, ≤8 s	28/18/2.197	shipped
48	MD3 shortest span 0.3 s	29/18/2.141	shipped (= B)
49	BANDLIST	0 hits, +6 wrong	rejected
51, 52b, 55	pause rules, kin-keep, frame-pause	415 STOP	rejected
52	SK7 strong keep	29/22/2.254	rejected
53, 53d	WEAK-WITNESS alone	27/13/2.113 (2 hits lost); 27/17/2.225	fail alone
53b, 53c	witness bars from the 415	outside allowed range	stop
54, 54b	SIGN (visible effect)	0 flips; “no” on 9/10	rejected
56	TWIN-SHORT	27/36/2.761	rejected
57	DEPICT-EVENT	29/17 (net 0 after cache fill)	shipped
<i>Rounds 58–66 (1–2 Oct)</i>			

round	idea	result	verdict
58, 58b	tight listener cut; FlexSED witness	415 precision 0.400, 0.297	stop
59	context ± 5 s	415 0.333	stop
60	SCENE-MARGIN + witness (= D)	29/14/2.028; TEST 24/23/2.386 (p 0.0145)	shipped
61b-d	look-alike reasoning (Qwen, Gemma)	hits lost in every variant	fail
62, 62b	logit gate; PRIOR	AUROC 0.647 / 0.644 < 0.65	stop
60L	scene check by logit margin (= the final system)	29/15/2.056; TEST 24/24/2.409 (p 0.034)	shipped, frozen
63	tagger ensemble	415 recall 0.152 vs 0.197	stop
64	concealed action {Bell}	30/14/1.972 (one clip)	not general, not shipped
65	perceptual-onset return	415 0.115	stop
66	NAME-ALL crop margin	gate screen passed; DEV 26/12/2.141 (3 hits lost)	fail

References

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